

Chapter 5

Reachability and controllability

In the previous chapters we mostly dealt with the behaviour of systems in unforced evolution. We shall now consider the problem of analysing to which extent input variables can influence the system's dynamics, and in order to do so we will introduce the fundamental concepts of reachability and controllability.

Reachability properties concern situations in which the initial state of a system is fixed, and the goal is to describe the set of states the system can be steered into by applying suitable admissible inputs. In controllability problems, on the other hand, the final state is fixed and the goal is to determine the set of initial states of the system for which there exist inputs that steer the system into the assigned final state.

The procedure to compute input functions that solve the problem of steering the system from initial state $\mathbf{x}'$ to final state $\mathbf{x}''$ is not unique. If a solution exists, one can consider several possible objectives, which correspond to different solution techniques with various levels of complexity and difficulty. The simplest situation corresponds to determining a control law that steers the system from $\mathbf{x}'$ to $\mathbf{x}''$ without imposing any additional constraint on how such evolution occurs. In this case there exists, in general, a set of equivalent solutions. If instead requirements are placed over the temporal evolution of the input function and the state variables, one must introduce criteria to choose within the set of solutions.

For linear systems reachability and controllability properties are usually referred to the state $\mathbf{x} = \mathbf{0}$, meaning that a state is said to be reachable, or controllable, with respect to the initial, or final, state $\mathbf{x} = \mathbf{0}$.

5.1 Reachability for linear discrete-time systems

Let $\Sigma = (F, G, H)$ be a linear discrete-time system. If the initial state is zero, $\mathbf{x}(0) = \mathbf{0}$, and if the applied input is

$$\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-1),$$

the state that is reached at time k is

$$\mathbf{x}(k) = \sum_{\sigma=0}^{k-1} F^{k-1-\sigma} G \mathbf{u}(\sigma) = [G \quad FG \quad \dots \quad F^{k-1}G] \begin{bmatrix} \mathbf{u}(k-1) \\ \mathbf{u}(k-2) \\ \vdots \\ \mathbf{u}(0) \end{bmatrix}. \quad (5.1)$$

As $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-1)$ varies, the set X_k^R of *reachable states in k steps* is the image, namely, the set of possible linear combinations of the columns, of the matrix

$$\mathcal{R}_k := [G \quad FG \quad \dots \quad F^{k-1}G],$$

that we call *reachability matrix in k steps*. It is clear that that

$$X_k^R = \text{Im} \mathcal{R}_k \quad (5.2)$$

is a vector subspace of the state space X , and as such it is called the *k -step reachable (sub)space*. If $X_k^R = X$, namely if $\text{rank} \mathcal{R}_k = n$, the system Σ is *reachable in k steps*.

Example 5.1.1 Consider the linear discrete-time system depicted in the diagram in Figure 5.1.1.

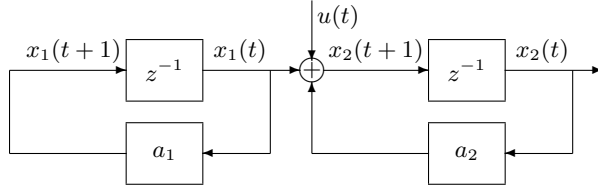


Figure 5.1.1

The state update equations are

$$\begin{aligned} x_1(t+1) &= a_1 x_1(t) \\ x_2(t+1) &= x_1(t) + a_2 x_2(t) + u(t) \end{aligned}$$

and, in matrix form,

$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \end{bmatrix} = \begin{bmatrix} a_1 & 0 \\ 1 & a_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ = F\mathbf{x}(t) + \mathbf{g}u(t).$$

One thus has, for any $k > 0$,

$$X_k^R = \text{Im} [\mathbf{g} \quad F\mathbf{g} \quad \dots \quad F^{k-1}\mathbf{g}] = \text{Im} \begin{bmatrix} 0 & 0 & \dots & 0 \\ 1 & a_2 & \dots & a_2^{k-1} \end{bmatrix} = \text{span} \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The result can be intuitively justified by observing that, if the initial state of the system is the zero state, the first delay element remains in the zero state, for any $u(\cdot)$.

If, instead, $u(\cdot)$ acts as input to the first delay, as in Figure 5.1.2, the equations become

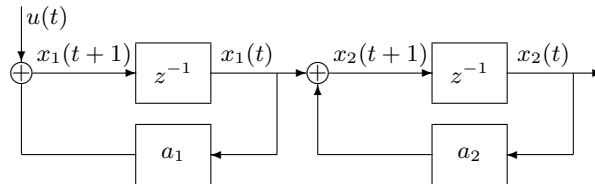


Figure 5.1.2

$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \end{bmatrix} = \begin{bmatrix} a_1 & 0 \\ 1 & a_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

and we obtain, for $k > 1$,

$$X_k^R = \text{Im} \begin{bmatrix} 1 & a_1 & \dots \\ 0 & 1 & \dots \end{bmatrix} = \mathbb{R}^2.$$

As a consequence, X_k^R coincides, for $k = 2, 3, \dots$, with the whole state space, and any state of the system can be reached starting from the zero state in at most two steps.

The subspaces reachable in $1, 2, \dots$ steps satisfy the chain of inclusions

$$X_1^R \subseteq X_2^R \subseteq \dots \subseteq X_\nu^R \subseteq X_{\nu+1}^R \subseteq \dots \quad (5.3)$$

Verifying condition (5.3) is straightforward, as the columns of the reachability matrix in ν steps are a subset of those of the reachability matrix in $\nu + 1$ steps.

Alternatively, one can observe that a state $\bar{\mathbf{x}}$, which is reachable from the zero state in ν steps by applying the input $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(\nu-1)$, is reachable in $\nu+1$ steps, by applying the input $\mathbf{v}(0) = \mathbf{0}, \mathbf{v}(1) = \mathbf{u}(0), \dots, \mathbf{v}(\nu) = \mathbf{u}(\nu-1)$.

Lemma 5.1.1 *If in the chain (5.3) two consecutive subspaces X_ν^R and $X_{\nu+1}^R$ coincide, then the following subspaces $X_{\nu+2}^R, X_{\nu+3}^R, \dots$ coincide with X_ν^R too.*

PROOF We will show that $X_\nu^R = X_{\nu+1}^R$ implies $X_{\nu+1}^R = X_{\nu+2}^R$. Due to (5.3), it is sufficient to verify that, if a state $\bar{\mathbf{x}}$ belongs to $X_{\nu+2}^R$, then it belongs to $X_{\nu+1}^R$.

If $\bar{\mathbf{x}} \in X_{\nu+2}^R$, there exists a sequence

$$\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(\nu+1)$$

that steers the system in $\nu+2$ steps from the state $\mathbf{x}(0) = \mathbf{0}$ to the state $\mathbf{x}(\nu+2) = \bar{\mathbf{x}}$. The state $\tilde{\mathbf{x}} := \mathbf{x}(\nu+1)$, which is reached at time $\nu+1$ by applying the subsequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(\nu)$, is, by definition, an element of $X_{\nu+1}^R$, and thus, by assumption, an element of X_ν^R .

There thus exists another input sequence, $\mathbf{v}(0), \mathbf{v}(1), \dots, \mathbf{v}(\nu-1)$, which steers the system in ν steps from the zero state to $\tilde{\mathbf{x}}$, and the sequence

$$\mathbf{v}(0), \mathbf{v}(1), \dots, \mathbf{v}(\nu-1), \mathbf{v}(\nu) = \mathbf{u}(\nu+1)$$

steers it in $\nu+1$ steps from the zero state to $\bar{\mathbf{x}}$. ■

- EXERCISE 5.1.1 [REFINEMENT OF LEMMA 5.1.1] If $\dim X_{\nu+1}^R - \dim X_\nu^R = \ell$, then $\dim X_{\nu+2}^R - \dim X_{\nu+1}^R \leq \ell$
 ‡ Hint: if $F^{\nu+1}\mathbf{g}_{i_1}, F^{\nu+1}\mathbf{g}_{i_2}, \dots, F^{\nu+1}\mathbf{g}_{i_k}$ are independent of the columns of $[G \ FG \dots F^\nu G]$, then $F^\nu\mathbf{g}_{i_1}, F^\nu\mathbf{g}_{i_2}, \dots, F^\nu\mathbf{g}_{i_k}$ are independent of the columns of $[G \ FG \dots F^{\nu-1}G]$.

Proposition 5.1.2 [REACHABLE SUBSPACE] *In a system of dimension n the chain of subspaces (5.3) is stationary (at least) starting from the n -th step, namely*

$$X_1^R \subseteq X_2^R \subseteq \dots \subseteq X_n^R = X_{n+1}^R = \dots$$

The subspace

$$X^R := X_n^R = X_{n+1}^R = \dots, \quad (5.4)$$

which is called the reachable (sub)space, is the set of states into which it is possible to steer the system starting from the zero state by applying suitable input sequences.

PROOF If the equality sign appears in the chain (5.3) anywhere previously to the subspace X_n^R , by applying Lemma 5.1.1 one concludes that $X_n^R = X_{n+1}^R = \dots$.

If all the inclusions until the space X_n^R are proper, then X_1^R is not the null space, as otherwise G would be the zero matrix and (5.3) would be stationary already at the first step, and the dimension of the subspaces $X_1^R, X_2^R, \dots, X_n^R$ is strictly monotonically increasing.

Furthermore, it necessarily holds true that $\dim X_1^R = 1$ and $\dim X_n^R = n$. Therefore, X_n^R coincides with the whole state space X , and the subspaces that follow the n -th subspace are all identical and equal to X_n^R . ■

It follows from Proposition 5.1.2 that in an n -dimensional system any state that is reachable can be reached in at most n steps. Nonetheless, the chain (5.3) can become stationary after $\bar{n} < n$ steps, and in such a case any reachable state can be reached in at most $\bar{n}$ steps.

From now on we shall denote by $\mathcal{R}$ the reachability matrix in n steps

$$\mathcal{R} = [G \quad FG \quad \dots F^{n-1}G]. \quad (5.5)$$

and we will refer to it as the *reachability matrix* of the system $\Sigma = (F, G, H)$, or, more concisely, of the pair (F, G) . The reachable subspace of the system is therefore

$$X^R = \text{Im} \mathcal{R} \quad (5.6)$$

and the system Σ is said to be *reachable* (or *completely reachable*) if $X^R = X$. A necessary and sufficient condition for the system to be reachable is

$$\text{rank} \mathcal{R} = n.$$

Example 5.1.2 [REACHABILITY OF SYSTEMS WITH SINGLE INPUT AND DIAGONAL F] In single input systems, the reachability condition can be expressed in a very simple form when the matrix F is diagonal. Consider in fact the system

$$\mathbf{x}(t+1) = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{bmatrix} u(t) = F\mathbf{x}(t) + \mathbf{g}u(t) \quad (5.7)$$

corresponding to the block diagram in Figure 5.1.3.

The reachability matrix

$$\begin{aligned} \mathcal{R} &= [\mathbf{g} \quad F\mathbf{g} \quad F^2\mathbf{g} \quad \dots \quad F^{n-1}\mathbf{g}] \\ &= \begin{bmatrix} g_1 & \lambda_1 g_1 & \lambda_1^2 g_1 & \dots & \lambda_1^{n-1} g_1 \\ g_2 & \lambda_2 g_2 & \lambda_2^2 g_2 & \dots & \lambda_2^{n-1} g_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_n & \lambda_n g_n & \lambda_n^2 g_n & \dots & \lambda_n^{n-1} g_n \end{bmatrix} \end{aligned}$$

is of full rank if and only if

1. $g_1, g_2, \dots, g_n$ are all different from zero, and
2. the eigenvalues of $\lambda_1, \lambda_2, \dots, \lambda_n$ are all distinct.

In order to establish such an equivalence, we factorise $\mathcal{R}$ into the product of two square matrices as follows

$$\mathcal{R} = \begin{bmatrix} g_1 & & & \\ & g_2 & & \\ & & \ddots & \\ & & & g_n \end{bmatrix} \begin{bmatrix} 1 & \lambda_1 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \dots & \lambda_n^{n-1} \end{bmatrix}. \quad (5.8)$$

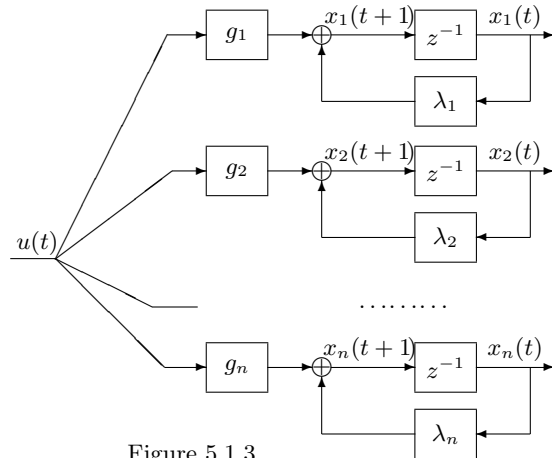


Figure 5.1.3

The matrix $\mathcal{R}$ has rank n if and only if both determinants of the two factors in (5.8) are different from zero. Therefore, the first condition follows from the diagonal structure of the first factor, while the second is due to the second factor being a Vandermonde matrix, which is invertible if and only $\lambda_i \neq \lambda_j$ if $i \neq j$.

This result has an evocative interpretation in terms of the diagram in Figure 5.1.3. In order for the system to be reachable, there must be an actual connection (namely, $g_i \neq 0$) between the input and each of the elementary subsystems in Figure 5.1.3, otherwise the state variables of the subsystems would remain identically zero. Furthermore, there cannot be two identical eigenvalues, otherwise the state variables of the corresponding subsystems would maintain, in time, a constant ratio.

Example 5.1.3* [CONTROL INPUT IN SYSTEMS WITH DIAGONAL F] If, in the previous example, the system is reachable, then any state $\bar{\mathbf{x}} \in \mathbb{R}^n$ can be reached at time $t = n$ by applying the (unique) input sequence $u(0), u(1), \dots, u(n-1)$ satisfying the vector equation

$$\bar{\mathbf{x}} = \mathcal{R} \begin{bmatrix} u(n-1) \\ u(n-2) \\ \vdots \\ u(0) \end{bmatrix} = \begin{bmatrix} g_1 & & & \\ & g_2 & & \\ & & \ddots & \\ & & & g_n \end{bmatrix} \begin{bmatrix} 1 & \lambda_1 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \dots & \lambda_n^{n-1} \end{bmatrix} \begin{bmatrix} u(n-1) \\ u(n-2) \\ \vdots \\ u(0) \end{bmatrix},$$

namely

$$\begin{bmatrix} g_1^{-1} & & & \\ & g_2^{-1} & & \\ & & \ddots & \\ & & & g_n^{-1} \end{bmatrix} \bar{\mathbf{x}} = \begin{bmatrix} 1 & \lambda_1 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \dots & \lambda_n^{n-1} \end{bmatrix} \begin{bmatrix} u(n-1) \\ u(n-2) \\ \vdots \\ u(0) \end{bmatrix}. \quad (5.9)$$

Therefore, the entries of $\bar{\mathbf{x}}$ and those of the input must solve the system

$$\begin{aligned} g_1^{-1} \bar{x}_1 &= u(n-1) + \lambda_1 u(n-2) + \dots + \lambda_1^{n-2} u(1) + \lambda_1^{n-1} u(0) \\ g_2^{-1} \bar{x}_2 &= u(n-1) + \lambda_2 u(n-2) + \dots + \lambda_2^{n-2} u(1) + \lambda_2^{n-1} u(0) \\ &\dots \dots \dots \\ g_n^{-1} \bar{x}_n &= u(n-1) + \lambda_n u(n-2) + \dots + \lambda_n^{n-2} u(1) + \lambda_n^{n-1} u(0). \end{aligned}$$

By multiplying by z^{n-1} the input representation $U(z^{-1}) = u(0) + u(1)z^{-1} + u(2)z^{-2} + \dots + u(n-1)z^{-(n-1)}$, we obtain the polynomial in the unknown z (whose degree is at most $n-1$)

$$\ell(z) := z^{n-1}U(z^{-1}) = u(0)z^{n-1} + u(1)z^{n-2} + \dots + u(n-2)z + u(n-1),$$

satisfying the following interpolation conditions

$$\ell(\lambda_1) = g_1^{-1} \bar{x}_1, \quad \ell(\lambda_2) = g_2^{-1} \bar{x}_2, \quad \dots, \quad \ell(\lambda_n) = g_n^{-1} \bar{x}_n.$$

In particular, if the assigned state to be reached is $\bar{\mathbf{x}} = g_k \mathbf{e}_k$, the corresponding polynomial $\ell_k(z)$ must be zero when z takes the values $\lambda_i \neq \lambda_k$, and therefore factorises as

$$\ell_k(z) = \gamma \prod_{\substack{i=1 \\ i \neq k}}^n (z - \lambda_i), \quad \gamma \in \mathbb{R},$$

and must take the value 1 when $z = \lambda_k$, and thus takes the form

$$\ell_k(z) = \prod_{\substack{i=1 \\ i \neq k}}^n (z - \lambda_i) \bigg/ \prod_{\substack{i=1 \\ i \neq k}}^n (\lambda_k - \lambda_i).$$

More in general, when $\bar{\mathbf{x}} = \sum_{k=1}^n \bar{x}_k \mathbf{e}_k$, by linearity we obtain

$$\ell(z) = \sum_{k=1}^n \frac{\bar{x}_k}{g_k} \left[\prod_{\substack{i=1 \\ i \neq k}}^n (z - \lambda_i) \bigg/ \prod_{\substack{i=1 \\ i \neq k}}^n (\lambda_k - \lambda_i) \right] \quad (5.10)$$

and finally

$$U(z^{-1}) = z^{-n+1}\ell(z) = \sum_{k=1}^n \frac{\bar{x}_k}{g_k} \left[\prod_{\substack{i=1 \\ i \neq k}}^n (1 - \lambda_i z^{-1}) \right] / \left[\prod_{\substack{i=1 \\ i \neq k}}^n (\lambda_k - \lambda_i) \right].$$

Equation (5.10) is in fact “Lagrange’s interpolating polynomial” formula of degree $n - 1$, whose graph $\mathbb{R}^2$ goes through the points $(\lambda_1, \frac{\bar{x}_1}{g_1}), \dots, (\lambda_n, \frac{\bar{x}_n}{g_n})$. Determining the input that steers the state of the system from $\mathbf{0}$ to $\bar{\mathbf{x}}$ in n steps can thus be cast into solving an interpolation problem.

- EXERCISE 5.1.2* [INVERSION OF A VANDERMONDE MATRIX] If $\lambda_i \neq \lambda_k$ for any $i \neq k$, determine the structure of the inverse of the Vandermonde matrix

$$V(\lambda_1, \dots, \lambda_n)^{-1} = \begin{bmatrix} 1 & \lambda_1 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \dots & \lambda_n^{n-1} \end{bmatrix}^{-1}.$$

‡ Hint: Setting $\mathbf{g} = [1 \ 1 \ \dots \ 1]^T$ and $\bar{\mathbf{x}} = \mathbf{e}_1$ in (5.9), one can observe that the first column of $V(\lambda_1, \dots, \lambda_n)^{-1}$ is the vector of the inputs $u(n-1), u(n-1), \dots, u(0)$ that steer the state to $\mathbf{x}(n) = \mathbf{e}_1$. The coefficients of the polynomial $\ell_1(z) = u(n-1) + u(n-2)z + \dots + u(0)z^{n-1} = \prod_{\substack{i=1 \\ i \neq 1}}^n (z - \lambda_i) / \prod_{\substack{i=1 \\ i \neq 1}}^n (\lambda_1 - \lambda_i)$ therefore provide, in the same order, the entries of the first column of $V(\lambda_1, \dots, \lambda_n)^{-1}$.

In a reachable system $\Sigma = (F, G, -)$, the least integer r for which

$$\text{rank} [G \quad FG \quad \dots \quad F^{r-1}G] = n \quad (5.11)$$

is called the *reachability index* of the system, and it coincides with the smallest index for which the subspaces in the chain (5.3) become equal to X .

A necessary and sufficient condition for the reachability index to be smaller than n is that a) the system has more than one input, and b) at least two inputs act independently, namely, that at least two columns of the matrix G are linearly independent. Necessity of such a condition is obvious. Sufficiency follows directly from Proposition 5.1.2, in fact, if

$$\dim X_1^R := \text{rank } G = \nu > 1,$$

and if the dimension of the spaces strictly increases by at least 1 in the chain (5.3), from its initial value ν up to n , then clearly the dimension n is obtained at latest by the space $X_{n-\nu+1}^R$.

- EXERCISE 5.1.3 Assume that (F, G) is reachable.
 - i) Condition $\text{rank } G = n$ is necessary and sufficient for the reachability index to be 1.
 - ii) Condition $\text{rank } G = 1$ is necessary and sufficient for the reachability index to be n .

The next result provides a geometrical characterization of the reachable subspace X^R .

Proposition 5.1.3 *Let $\Sigma = (F, G, H)$. The reachability subspace X^R is the smallest F -invariant subspace of X that includes $\text{Im } G$.*

PROOF It is obvious that $X^R = \text{Im} [G \quad FG \quad \dots \quad F^{n-1}G]$ includes $\text{Im } G$.

If $\mathbf{x} \in X^R$, then $F\mathbf{x}$ belongs to the image of the matrix $[FG \ F^2G \ \dots \ F^nG]$ and hence to the space X^R , since F^n is a linear combination of $I, F, \dots, F^{n-1}$ by Cayley-Hamilton's theorem. Therefore, X^R is F -invariant.

To show that X^R is the smallest subspace of X for which such a property holds, assume that S is a generic F -invariant subspace such that $S \supseteq \text{Im}G$. Then S includes $\text{Im} F^kG$ for any $k > 0$, and hence $S \supseteq X^R$. ■

By extending the notation adopted for cyclic spaces, the smallest F -invariant subspace that includes (the columns of) G will be denoted by $\langle F|G \rangle$. The previous result thus proves that

$$X^R = \langle F|G \rangle. \quad (5.12)$$

Remark Algebraically equivalent systems exhibit the same reachability properties. In fact, consider the algebraically equivalent systems $\Sigma = (F, G, H)$ and $\bar{\Sigma} = (\bar{F}, \bar{G}, \bar{H})$, and let T be the change of basis matrix such that

$$\bar{F} = T^{-1}FT, \quad \bar{G} = T^{-1}G, \quad \bar{H} = HT.$$

The k -steps reachable subspaces of Σ and $\bar{\Sigma}$ are related by

$$\bar{X}_k^R = \text{Im} [\bar{G} \ \bar{F}\bar{G} \ \dots \ \bar{F}^{k-1}\bar{G}] = \text{Im} (T^{-1} [G \ FG \ \dots \ F^{k-1}G]) = T^{-1}X_k^R \quad (5.13)$$

and have the same dimension. Notice that (11.6) yields the elements of $\bar{X}_k^R$ as transformed versions of the corresponding elements of X_k^R .

Denoting by $\mathcal{R}$ and $\bar{\mathcal{R}}$ the reachability matrices of Σ and $\bar{\Sigma}$, respectively, from (11.6), and for $k = n$, one obtains

$$\bar{\mathcal{R}} = T^{-1}\mathcal{R}. \quad (5.14)$$

If the systems are reachable, then by means of (5.14) it is possible to uniquely determine the change of basis matrix T whenever both $\mathcal{R}$ and $\bar{\mathcal{R}}$ are known. In fact, by post-multiplying both sides of (5.14) by $\bar{\mathcal{R}}^T$, one obtains

$$\bar{\mathcal{R}}\bar{\mathcal{R}}^T = T^{-1}\mathcal{R}\bar{\mathcal{R}}^T$$

Since $\bar{\mathcal{R}}\bar{\mathcal{R}}^T$ is invertible, the last equality implies

$$T = \mathcal{R}\bar{\mathcal{R}}^T(\bar{\mathcal{R}}\bar{\mathcal{R}}^T)^{-1}. \quad (5.15)$$

In particular, if the systems only have a single input, then $\mathcal{R}$ and $\bar{\mathcal{R}}$ are square and non-singular, and from (5.14) it directly follows

$$T = \mathcal{R}\bar{\mathcal{R}}^{-1}. \quad (5.16)$$

5.2 Controllability of linear discrete-time systems

The results derived in the previous paragraph for the study of the reachability properties can be exploited to address the controllability problem.

Definition 5.2.1 The state $\mathbf{x}$ of the discrete-time system $\Sigma = (F, G, H)$ is *controllable (to zero) in k steps* if there exists an input sequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-1)$ steering the system from the state $\mathbf{x}$ to the state $\mathbf{0}$ over the interval $[0, k]$, namely if $\mathbf{0} = F^k \mathbf{x} + \sum_{\sigma=0}^{k-1} F^{k-1-\sigma} G \mathbf{u}(\sigma)$, or, equivalently, if

$$-F^k \mathbf{x} = \sum_{\sigma=0}^{k-1} F^{k-1-\sigma} G \mathbf{u}(\sigma). \quad (5.17)$$

Notice that (5.17) can be interpreted as the condition of k -steps reachability for the state $-F^k \mathbf{x}$. Therefore, it can be rewritten as

$$F^k \mathbf{x} \in \text{Im} [G \quad FG \quad \dots \quad F^{k-1}G] = X_k^R. \quad (5.18)$$

The set X_k^C of states that satisfy (5.18),

$$X_k^C := \{\mathbf{x} \in X : F^k \mathbf{x} \in X_k^R\},$$

has the structure of a vector space and is thus called the *k -steps controllable subspace*. The discrete-time system (F, G, H) is *controllable in k steps* if (5.18) holds true for any $\mathbf{x} \in X$, namely if the image of X through F^k is included in X_k^R

$$\text{Im} F^k \subseteq \text{Im} [G \quad FG \quad \dots \quad F^{k-1}G] = X_k^R. \quad (5.19)$$

Example 5.2.1 Consider the discrete-time system with matrices

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The one-step controllable subspace consists of

$$X_1^C = \{\mathbf{x} : F\mathbf{x} \in \text{Im}[\mathbf{g}]\} = \left\{ \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} : \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \in \text{Im} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} \alpha \\ 0 \\ \gamma \end{bmatrix}, \alpha, \gamma \in \mathbb{R} \right\},$$

whereas the two-steps controllable subspace is

$$X_2^C = \{\mathbf{x} : F^2 \mathbf{x} \in \text{Im}[\mathbf{g} \quad F\mathbf{g}]\} = \left\{ \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} : \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \in \text{Im} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \right\} = X.$$

Therefore, the system is controllable in two steps.

- EXERCISE 5.2.1 (i) $X_1^C = \{\mathbf{0}\}$ if and only if F is invertible and G is the zero matrix.
 ii) $X_1^C = \{\mathbf{0}\}$ implies $X_k^C = \{\mathbf{0}\}$ for all $k > 0$.
 ‡ *Solution:* (i) $X_1^C = \{\mathbf{x} : F\mathbf{x} \in \text{Im}G\} = \{\mathbf{0}\}$ implies $\ker F = \{\mathbf{0}\}$, thus F is invertible. This in turn implies $\{\mathbf{x} : F\mathbf{x} \in \text{Im}G\} = \{\mathbf{x} : \mathbf{x} \in \text{Im} F^{-1}G\}$, and G must be the zero matrix. The converse is straightforward since if F is invertible and G is the zero matrix then the subspace X_1^C only contains the zero vector.
 (ii) Since F is invertible, the same holds true for F^k . At the same time, $G = 0$ implies $X_k^R = \{\mathbf{0}\}$. Then, the condition $F^k \mathbf{x} \in X_k^R$ is satisfied only by $\mathbf{x} = \mathbf{0}$.

Any state that is controllable in k steps is also controllable in $k + 1$ steps. Let in fact $\bar{\mathbf{x}}$ be controllable to zero in k steps. Then there exists an input sequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-1)$ that steers the system from the state $\mathbf{x}(0) = \bar{\mathbf{x}}$ to the state $\mathbf{x}(k) = \mathbf{0}$. Then obviously applying $\mathbf{u}(k) = \mathbf{0}$ the system remains in the zero state at time $k + 1$.

Since the previous observation holds true for any k , it proves the following chain of inclusions of subspaces of X

$$X_1^C \subseteq X_2^C \subseteq \dots \subseteq X_\nu^C \subseteq X_{\nu+1}^C \subseteq \dots \quad (5.20)$$

Lemma 5.2.2 *If in the chain (5.20) two consecutive subspaces X_ν^C and $X_{\nu+1}^C$ coincide, then all the subsequent $X_{\nu+2}^C, X_{\nu+3}^C, \dots$ coincide with X_ν^C .*

PROOF We will show that $X_\nu^C = X_{\nu+1}^C$ implies $X_{\nu+1}^C = X_{\nu+2}^C$. In order to do so, it suffices to verify that any state $\bar{\mathbf{x}} \in X_{\nu+2}^C$ also belongs to $X_{\nu+1}^C$.

In fact, if $\bar{\mathbf{x}} \in X_{\nu+2}^C$, then there exists a sequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(\nu+1)$ that steers the system from the state $\mathbf{x}(0) = \bar{\mathbf{x}}$ to the state $\mathbf{x}(\nu+2) = \mathbf{0}$ in $\nu+2$ steps.

Obviously, the sequence $\mathbf{u}(1), \mathbf{u}(2), \dots, \mathbf{u}(\nu+1)$ steers the system from the state $\mathbf{x}(1) = F\bar{\mathbf{x}} + G\mathbf{u}(0)$ to the state $\mathbf{x}(\nu+2) = \mathbf{0}$ in $\nu+1$ steps, thus $\mathbf{x}(1) \in X_{\nu+1}^C$.

Since by assumption $X_\nu^C = X_{\nu+1}^C$, condition $\mathbf{x}(1) \in X_{\nu+1}^C$ implies $\mathbf{x}(1) \in X_\nu^C$, therefore there exists another input sequence

$$\mathbf{v}(1), \mathbf{v}(2), \dots, \mathbf{v}(\nu)$$

that steers the system from the state $\mathbf{x}(1) = F\bar{\mathbf{x}} + G\mathbf{u}(0)$ to the state $\mathbf{x}(\nu+1) = \mathbf{0}$ in ν steps. Then, as required,

$$\mathbf{u}(0), \mathbf{v}(1), \mathbf{v}(2), \dots, \mathbf{v}(\nu)$$

steers the system from the state $\bar{\mathbf{x}}$ to the state $\mathbf{0}$ in $\nu+1$ steps. ■

Proposition 5.2.3 *In an n -dimensional system the chain of subspaces (5.20) is stationary at least when it reaches the n -th step (if not earlier), namely*

$$X_1^C \subseteq X_2^C \subseteq \dots \subseteq X_n^C = X_{n+1}^C = \dots$$

The controllable subspace

$$X^C := X_n^C = \{\mathbf{x} : F^n \mathbf{x} \in X^R\} = X_{n+1}^C = \dots \quad (5.21)$$

represents the set of initial states that are controllable to zero, namely, the set of states for which there exists an input sequence that steers the system to the zero state in an interval of time of suitable duration.

PROOF If at any point preceding X_n^C the chain exhibits the equality sign, then by Lemma 5.2.2 one has $X_n^C = X_{n+1}^C = \dots$.

If the inclusions until the space X_n^C are all proper, then X_1^C is not the null space, otherwise X_2^C would also be the null space (see Exercise 5.2.1), and we would have $X_1^C = X_2^C$. Then, the dimensions of the subspaces $X_1^C \subset X_2^C \subset \dots \subset X_n^C$ are strictly increasing and it necessarily holds true that $\dim X_1^C = 1$ and $\dim X_n^C = n$. Therefore, $X_n^C = X$ and the subsequent subspaces are equal to X_n^C . ■

System $\Sigma = (F, G, H)$ is *controllable* if $X^C = X$. Due to the proposition above, this condition holds true if and only if the entire state space $X = \mathbb{R}^n$ is controllable in k steps, for some value k , or, more simply, if it is controllable in n steps. Then from

$$X^C = X_n^C = \{\mathbf{x} : F^n \mathbf{x} \in X^R\}$$

one can easily prove the next result.

Proposition 5.2.4 [CONTROLLABILITY CONDITION] *System $\Sigma = (F, G, -)$ is controllable if and only if*

$$\text{Im} F^n \subseteq X^R = \text{Im} [G \quad FG \quad \dots \quad F^{n-1}G]. \quad (5.22)$$

For discrete-time systems, reachability and controllability in k steps –and, in particular, in n steps– are not equivalent properties. In fact, while the former implies the latter, the converse is not true, in general. To establish the first claim, it suffices to notice that if the system is reachable in k steps, then $X_k^R = X$, thus the controllability condition in k steps, $\text{Im} F^k \subseteq X_k^R$, is trivially satisfied. Vice versa, any system for which F is the zero matrix is controllable in k steps, for any $k > 0$, but it is not reachable in k steps, unless the rank of G is equal to n .

- EXERCISE 5.2.2 If F is nilpotent, $\Sigma = (F, G, H)$ is always controllable, but its reachability depends on the choice of G . For example, the system with

$$F = \begin{bmatrix} 0 & 1 & & & \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \dots & \dots & 1 \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}$$

is controllable, but not reachable.

- EXERCISE 5.2.3 If F is invertible, the discrete-time system $\Sigma = (F, G, H)$ is controllable in k steps if and only if it is reachable in k steps. For example

$$F = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is not reachable and, since the matrix F is invertible, neither is controllable.

Proposition 5.2.5 [GEOMETRIC PROPERTIES OF THE CONTROLLABLE SUBSPACE] *The controllable subspace X^C is F -invariant and includes $\text{Im} G$. Therefore,*

$$X^C \supseteq X^R. \quad (5.23)$$

PROOF If $\mathbf{x} \in X^C$, then $F^n \mathbf{x} \in X^R$, and thus $F^n(F\mathbf{x}) = F(F^n \mathbf{x}) \in FX^R \subseteq X^R$, due to X^R being F -invariant. Then, the same holds true for $F\mathbf{x} \in X^C$. Moreover from $\text{Im} G \subseteq X^R$ and by the F -invariance of X^R it follows that the columns of $F^n G$ are elements of X^R . Hence, $\text{Im} G \subseteq X^C$. Since X^R is the smallest F -invariant subspace that includes $\text{Im} G$, the subspace X^C includes X^R . ■

- EXERCISE 5.2.4* X^C is the (not necessarily direct) sum of X^R and the (possibly generalised) eigenspace of F corresponding to the eigenvalue 0. Consequently X^C contains X^R and is F -invariant.
 $\sharp$ *Solution.* The generalised eigenspace of F corresponding to the eigenvalue 0 is the kernel of F^n . We thus have to prove that $\mathbf{x} \in X^C$ if and only if it is the sum of two vectors, $\mathbf{r}$ and $\mathbf{s}$, such that $\mathbf{r} \in X^R$ and $\mathbf{s} \in \ker F^n$.
 If $\mathbf{x} = \mathbf{r} + \mathbf{s}$, with $\mathbf{r} \in X^R$ and $F^n \mathbf{s} = \mathbf{0}$, then $F^n \mathbf{x} = F^n \mathbf{r} + F^n \mathbf{s} = F^n \mathbf{r} \in X^R$, since X^R is F -invariant. Therefore, $\mathbf{x} \in X^C$.
 Conversely, assume $\mathbf{x} \in X^C$, which implies $F^n \mathbf{x} \in X^R$. Let $z^k + \eta_{k+1}z^{k+1} + \dots + \eta_{k+m}z^{k+m}$ be an annihilating polynomial of the vector $F^n \mathbf{x}$ with respect to the matrix F^n . One has

$$\begin{aligned} \mathbf{0} &= F^{kn}(F^n \mathbf{x}) + \eta_{k+1}F^{(k+1)n}(F^n \mathbf{x}) + \dots + \eta_{k+m}F^{(k+m)n}(F^n \mathbf{x}) \\ &= F^{(k+1)n}[\mathbf{x} + \eta_{k+1}F^n \mathbf{x} + \dots + \eta_{k+m}F^{mn} \mathbf{x}] \end{aligned}$$

and the conclusion then follows by setting $\mathbf{x} = \mathbf{r} + \mathbf{s}$ with

$$\begin{aligned} -\mathbf{r} &:= \eta_{k+1}F^n \mathbf{x} + \dots + \eta_{k+m}F^{mn} \mathbf{x} \in X^R \\ \mathbf{s} &:= \mathbf{x} + \eta_{k+1}F^n \mathbf{x} + \dots + \eta_{k+m}F^{mn} \mathbf{x} \in \ker F^{(k+1)n} = \ker F^n. \end{aligned}$$

- EXERCISE 5.2.5 Let $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(n-1)$ and $\mathbf{v}(0), \mathbf{v}(1), \dots, \mathbf{v}(n-1)$ be two input sequences that control to zero respectively the states $\mathbf{x}$ and the state $\bar{\mathbf{x}} = G\mathbf{u}(0)$. Determine an input sequence that controls to zero the state $F\mathbf{x}$.

In general, the inclusion $X_n^R \subseteq X_n^C$ does not extend to k -steps reachability and controllability subspaces, when $k < n$. However, the following result holds true.

Proposition 5.2.6 [DIMENSIONS OF THE SUBSPACES X_k^R AND X_k^C] For any $k > 0$ the subspace X_k^R has dimension smaller than or equal to that of X_k^C .

PROOF* If X_k^R has dimension ρ , let $B_k \in \mathbb{R}^{n \times \rho}$ be a matrix whose columns constitute a basis for X_k^R . Then

$$X_k^C = \{\mathbf{x} : F^k \mathbf{x} \in \text{Im } B_k\} = \{\mathbf{x} : F^k \mathbf{x} = -B_k \mathbf{w}, \exists \mathbf{w} \in \mathbb{R}^\rho\} = \{\mathbf{x} : \begin{bmatrix} F^k & B_k \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{w} \end{bmatrix} = \mathbf{0}, \exists \mathbf{w}\}.$$

We shall denote by π the projection map $\pi : \mathbb{R}^{n+\rho} \rightarrow \mathbb{R}^n : \begin{bmatrix} \mathbf{x} \\ \mathbf{w} \end{bmatrix} \mapsto \mathbf{x}$. The previous formula implies that the k -steps controllable space X_k^C coincides with the image through π of the kernel of the matrix $\begin{bmatrix} F^k & B_k \end{bmatrix}$, namely

$$X_k^C = \pi \left(\ker \begin{bmatrix} F^k & B_k \end{bmatrix} \right). \quad (5.24)$$

If two vectors $\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{w}_1 \end{bmatrix}$ and $\begin{bmatrix} \mathbf{x}_2 \\ \mathbf{w}_2 \end{bmatrix}$ in $\ker \begin{bmatrix} F^k & B_k \end{bmatrix}$ have the same projection, then it must hold true $\mathbf{x}_1 = \mathbf{x}_2$, and therefore $\begin{bmatrix} \mathbf{0} \\ \mathbf{w}_1 - \mathbf{w}_2 \end{bmatrix} \in \ker \begin{bmatrix} F^k & B_k \end{bmatrix}$, so in turn $B_k(\mathbf{w}_1 - \mathbf{w}_2) = \mathbf{0}$, and finally $\mathbf{w}_1 = \mathbf{w}_2$, because the columns of B_k are linearly independent. This proves that π is an injective map when it is applied to the kernel of $\begin{bmatrix} F^k & B_k \end{bmatrix}$. Therefore, one has

$$\begin{aligned} \dim X_k^C &= \dim \left\{ \pi \left(\ker \begin{bmatrix} F^k & B_k \end{bmatrix} \right) \right\} = \dim \ker \begin{bmatrix} F^k & B_k \end{bmatrix} \\ &= \{n + \rho\} - \dim \text{Im} \begin{bmatrix} F^k & B_k \end{bmatrix} = \rho + \{n - \dim \text{Im} \begin{bmatrix} F^k & B_k \end{bmatrix}\} \geq \rho. \end{aligned} \quad \blacksquare$$

Example 5.2.2 Consider the discrete-time system with matrices

$$F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

The 1-step reachable subspace is

$$X_1^R = \text{Im } \mathbf{g} = \text{span} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

while all subspaces that are reachable in at least two steps are the same, and equal to

$$X_k^R = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad k = 2, 3, \dots$$

The 1-step controllable subspace

$$X_1^C = \left\{ \mathbf{x} : F\mathbf{x} \in \text{span} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} : \begin{bmatrix} 0 \\ 0 \\ \gamma \end{bmatrix} \in \text{span} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} \alpha \\ \beta \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$$

has dimension 2, but does not contain X_1^R . Verify that $X_2^C = X$, which implies that the system is controllable in two steps.

- **EXERCISE 5.2.5** Determine the 2-steps reachable and controllable subspaces of the discrete-time system

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Verify that, while their dimensions are the same, they do not coincide.

- **EXERCISE 5.2.6** If $X^C = X_k^C$, namely, if the chain of controllable subspaces becomes stationary after k steps, show that $X^R = X_k^R$. Does the converse hold true?

‡ *Hint: If by contradiction $X_k^R \subsetneq X_{k+1}^R$, then $\text{Im} F^k \mathbf{g} \not\subseteq X_k^R$. This implies that $X^C = X_k^C = \{\mathbf{x} : F^k \mathbf{x} \in X_k^R\}$ does not contain $\text{Im} \mathbf{g}$, which is a contradiction since by (5.23) it follows $X^C \supseteq X^R \supseteq \text{Im} \mathbf{g}$.*

5.3 Reachability and controllability of linear continuous-time systems

The study of reachability of linear continuous-time systems is conceptually analogous to that of discrete-time systems. The main difference lies in the type of analytical techniques that need to be employed, which differ from those adopted in the discrete-time case due to a) the dimension of the space of input functions, which is infinite, and b) the impossibility to provide a direct matrix representation of the relation between input function and reached state. As it is well known, for an n -dimensional, continuous-time system $\Sigma = (F, G, H)$, whose state equation is

$$\dot{\mathbf{x}}(t) = F\mathbf{x}(t) + G\mathbf{u}(t) \tag{5.25}$$

the state reached at time t in forced evolution and corresponding to input function $\mathbf{u}(\cdot)$ can be expressed by means of the formula of variation of constants

$$\mathbf{x}(t) = \int_0^t e^{F(t-\sigma)} G \mathbf{u}(\sigma) d\sigma. \quad (5.26)$$

Consequently, a state $\mathbf{x}$ is *reachable at time t* if there exists a suitable input function $\mathbf{u}(\cdot)$ such that the equality

$$\mathbf{x} = \int_0^t e^{F(t-\sigma)} G \mathbf{u}(\sigma) d\sigma$$

is satisfied.

In other terms, once we denote by $\mathcal{U}_{[0,t]}$ the space of input functions over the interval $[0, t]$, and we introduce the linear operator R_t that is defined as

$$R_t : \mathcal{U}_{[0,t]} \rightarrow X : \mathbf{u}(\cdot) \mapsto \int_0^t e^{F(t-\sigma)} G \mathbf{u}(\sigma) d\sigma, \quad (5.27)$$

then $\mathbf{x}$ is reachable at time t if and only if $\mathbf{x}$ belongs to the image of R_t .

For any $t > 0$ the *reachable subspace at time t* , X_t^R , can be thus characterised as the image of the linear operator R_t acting over the infinite dimensional space $\mathcal{U}_{[0,t]}$ into the finite-dimensional space $X = \mathbb{R}^n$, namely, $X_t^R = \text{Im} R_t$.

Recalling the properties of adjoint operators (see Appendix, section A.13), one can obtain a representation of X_t^R as the image of a matrix. In fact, the adjoint operator R_t^* of R_t is

$$R_t^* : X \rightarrow \mathcal{U}_{[0,t]} : \mathbf{x} \mapsto G^T e^{F^T(t-\sigma)} \mathbf{x}, \quad \sigma \in [0, t], \quad (5.28)$$

and since the dimension of $\text{Im} R_t \subseteq X$ is finite, the identity

$$\text{Im} R_t = \text{Im}(R_t R_t^*). \quad (5.29)$$

holds true. Then, the operator

$$\begin{array}{c} R_t R_t^* \\ R_t^* \end{array} : X \rightarrow X : \mathbf{x} \mapsto \begin{array}{c} G^T e^{F^T(t-\sigma)} \mathbf{x}, \sigma \in [0, t] \\ \int_0^t e^{F(t-\sigma)} G G^T e^{F^T(t-\sigma)} \mathbf{x} d\sigma \end{array} \quad (5.30)$$

maps the vector $\mathbf{x}$ into the vector $\mathcal{W}_t \mathbf{x}$, where

$$\mathcal{W}_t := \int_0^t e^{F(t-\sigma)} G G^T e^{F^T(t-\sigma)} d\sigma = \int_0^t e^{F\tau} G G^T e^{F^T\tau} d\tau \quad (5.31)$$

The matrix $\mathcal{W}_t$ is called the *reachability Gramian matrix* corresponding to the interval $[0, t]$. In such a way, the reachability subspace at time t can be expressed as the image of the $n \times n$ matrix $\mathcal{W}_t$, namely,

$$X_t^R = \text{Im} \mathcal{W}_t. \quad (5.32)$$

Differently from discrete-time systems, the reachability subspace for continuous-time systems does not depend on the duration of the interval $[0, t]$ over which the input function acts. This property is formally stated and proved in the following result.

Proposition 5.3.1 *In an n -dimensional continuous-time system $\Sigma = (F, G, H)$ the reachable subspace at time $t > 0$ is the image of the reachability matrix*

$$\mathcal{R} = [G \quad FG \quad \dots \quad F^{n-1}G].$$

System Σ is reachable if $\mathcal{R}$ has rank equal to n .

PROOF By the properties of the adjoint operator, one has

$$\text{Im} R_t = (\ker R_t^*)^\perp \quad \text{and} \quad \text{Im} \mathcal{R} = (\ker \mathcal{R}^T)^\perp,$$

therefore in order to show that the reachable subspace $\text{Im} R_t$ coincides with $\text{Im} \mathcal{R}$ it suffices to show that $\ker R_t^* = \ker \mathcal{R}^T$. In turn, the latter follows from the following chain of equivalent conditions

$$\mathbf{x} \in \ker R_t^* \Leftrightarrow 0 = G^T e^{F^T(t-\sigma)} \mathbf{x}, \quad \forall \sigma \in [0, t] \quad (5.33)$$

$$\Leftrightarrow 0 = \sum_{i=0}^{\infty} G^T (F^T)^i \mathbf{x} \frac{(t-\sigma)^i}{i!}, \quad \forall \sigma \in [0, t] \quad (5.34)$$

$$\Leftrightarrow 0 = G^T (F^T)^i \mathbf{x}, \quad i = 0, 1, 2, \dots \quad (5.35)$$

$$\Leftrightarrow 0 = G^T (F^T)^i \mathbf{x}, \quad i = 0, 1, \dots, n-1 \quad (5.36)$$

$$\Leftrightarrow \mathbf{x} \in \ker \begin{bmatrix} G^T \\ G^T F^T \\ \vdots \\ G^T (F^T)^{n-1} \end{bmatrix} = \ker \mathcal{R}^T. \quad (5.37)$$

In particular, the equivalence of (5.34) and (5.35) holds true due to the identity principle of the power series, and the equivalence of (5.35) and (5.36) follows from Cayley-Hamilton's theorem. ■

Since, for continuous-time systems, the reachable subspace is independent of t , it will be denoted by X^R from now on.

It follows from the previous analysis that the reachable subspace X^R is the image of the reachability matrix $\mathcal{R}$ both for continuous-time and discrete-time systems. Therefore, the properties of X^R that follow from the structure of $\mathcal{R}$, in particular the characterisation in Proposition 5.1.3, hold true for both classes of systems.

Remark The reachability of $\Sigma = (F, G, H)$ corresponds to $\ker \mathcal{R}^T = \{0\}$. By the proof of Proposition 5.3.1, and in particular by (5.35), the latter condition is also equivalent to the fact that $\mathbf{x}^T e^{F\sigma} G = 0$, $\forall \sigma \in [0, t]$, implies $\mathbf{x} = \mathbf{0}$. In turn, this is equivalent to the linear independence of the n rows, which are functions of σ , of the matrix $e^{F\sigma} G$.

- EXERCISE 5.3.1 The (i, j) -entry of the matrix $e^{Ft} G$ represents the time evolution of the i -th component of $\mathbf{x}_f(\cdot)$ when the unitary Dirac delta function is applied at the j -th input at time $t = 0$.

The controllability problem for continuous-time systems is defined in the same way as for linear discrete-time systems. In the present case, however, the solution is much simpler as the controllability subspace coincides with the reachability subspace.

To be more precise, a state $\bar{\mathbf{x}}$ is said to be controllable (to zero) at time $t > 0$ if there exists an input in $[0, t]$ that steers the systems from state $\bar{\mathbf{x}}$ to state $\mathbf{0}$, namely if

$$\mathbf{0} = e^{Ft} \bar{\mathbf{x}} + \int_0^t e^{F(t-\sigma)} G \mathbf{u}(\sigma) d\sigma \quad (5.38)$$

for a suitable choice of the input function $\mathbf{u}(\cdot)$. Since the integral in (5.38) yields, as $\mathbf{u}(\cdot)$ varies over the set of input functions, all the reachable states, namely, all the states in the reachability subspace, the controllability condition of $\bar{\mathbf{x}}$ over $[0, t]$ can be formulated as

$$e^{Ft}\bar{\mathbf{x}} \in X^R$$

namely

$$\bar{\mathbf{x}} \in e^{-Ft}X^R. \quad (5.39)$$

The set of states that are controllable at time t is therefore the following subspace of X

$$X_t^C = e^{-Ft}X^R, \quad (5.40)$$

We notice that X_t^C 1) is contained in X^R , because X^R is F -invariant and therefore also e^{-Ft} -invariant, and 2) has the same dimension as X^R , because the matrix e^{-Ft} is invertible. Then, it coincides with X^R . In particular, X_t^C does not depend on $t > 0$.

Due to the previous discussion, we will not distinguish between reachable and controllable subspaces for continuous-time systems.

- **EXERCISE 5.3.2** Show that for a continuous-time system $\Sigma = (F, G, -)$ the state $\mathbf{x}$ is controllable over $[0, t]$ if and only if it can be expressed in the form $\int_0^t e^{-F\sigma}G\mathbf{u}(\sigma)d\sigma$ for a suitable input function $\mathbf{u}(\cdot)$.

Example 5.3.1 Consider the electrical networks in Figure 5.3.1, where the input is the current $j(\cdot)$ and the output is the voltage $v(\cdot)$. We set

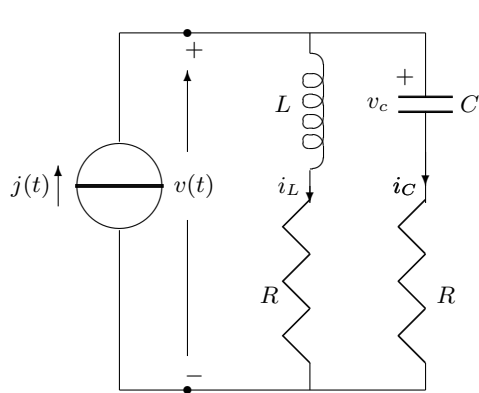


Figure 5.3.1

$$\mathbf{x} = \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

and from the equations

$$\begin{aligned} C \frac{dv_C}{dt} &= j - i_L \\ L \frac{di_L}{dt} &= v_C + Ri_C - Ri_L \\ &= v_C + R(j - i_L) - Ri_L \\ &= v_C - 2Ri_L + Rj \\ v &= v_C + Ri_C = v_C + R(j - i_L) \\ &= v_C - Ri_L + Rj \end{aligned}$$

we obtain the state-space model

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} -\frac{2R}{L} & \frac{1}{L} \\ -\frac{1}{C} & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} \frac{R}{L} \\ \frac{1}{C} \end{bmatrix} j(t) \\ v(t) &= \begin{bmatrix} -R & 1 \end{bmatrix} \mathbf{x}(t) + Rj(t). \end{aligned} \quad (5.41)$$

The reachability matrix, which is

$$\mathcal{R} = \begin{bmatrix} \frac{R}{L} & -\frac{2R^2}{L^2} + \frac{1}{LC} \\ \frac{1}{C} & -\frac{R}{LC} \end{bmatrix},$$

is of full rank except when $R^2 = L/C$, that is, when the time constants RC and L/R of the two branches in parallel are the same. The space X^R is generated by the columns of $\mathcal{R}$. In particular,

when $R^2 = L/C$, the space X^R can be written as

$$X^R = \text{Im}\mathcal{R} = \text{span} \begin{bmatrix} \frac{R}{L} \\ 1 \\ \frac{1}{C} \end{bmatrix}.$$

5.4 Control inputs and Gramian matrices

In this paragraph we will investigate the existence of, and if possible determine, the input $\mathbf{u}(\cdot)$ that steers, over an assigned and finite time horizon, the state of the system $\Sigma = (F, G, H)$ from an initial value $\mathbf{x}_0$ to a final value $\mathbf{x}_f$, both given.

We start by tackling the problem for linear discrete-time systems over the time horizon $[0, k]$. It is easy to see that there exists a sequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-1)$ that steers the system from the initial state $\mathbf{x}(0) = \mathbf{x}_0$ to the final state $\mathbf{x}(k) = \mathbf{x}_f$ if and only if

$$\mathbf{x}_f - F^k \mathbf{x}_0 \in \text{Im} [G \quad FG \quad \dots \quad F^{k-1}G] = X_k^R, \quad (5.42)$$

namely if and only if $\mathbf{x}_f - F^k \mathbf{x}_0$ is reachable, from the zero state, in k steps. Consequently, the states that can be reached from $\mathbf{x}_0$ in k steps constitute the set

$$F^k \mathbf{x}_0 + X_k^R,$$

which is an element of the quotient space X/X_k^R .

Assuming that $\mathbf{x}_0$ and $\mathbf{x}_f$ satisfy (5.42), in order to determine an input sequence we must solve the equation

$$\mathbf{x}_f - F^k \mathbf{x}_0 = \mathcal{R}_k \mathbf{u} \quad (5.43)$$

where the unknown is the vector

$$\mathbf{u} = \begin{bmatrix} \mathbf{u}(k-1) \\ \mathbf{u}(k-2) \\ \vdots \\ \mathbf{u}(0) \end{bmatrix} \in \mathbb{R}^{km}.$$

Such an equation corresponds to a linear system with n equations in km unknown, having as constant term the vector $\mathbf{x}_f - F^k \mathbf{x}_0$. Its general solution is the sum of a specific solution of (5.43) and of an arbitrary solution of the associated homogeneous system

$$\mathbf{0} = \mathcal{R}_k \mathbf{v}. \quad (5.44)$$

To obtain a specific solution $\bar{\mathbf{u}}$ to (5.43), one possible procedure is

- introduce the *reachability gramian in k steps* $\mathcal{W}_k := \mathcal{R}_k \mathcal{R}_k^T$,
- solve the auxiliary equation in the unknown $\eta \in \mathbb{R}^n$

$$\mathcal{W}_k \eta = (\mathcal{R}_k \mathcal{R}_k^T) \eta = \mathbf{x}_f - F^k \mathbf{x}_0 \quad (5.45)$$

- set

$$\bar{\mathbf{u}} = \mathcal{R}_k^T \eta. \quad (5.46)$$

Notice that (5.45) can be solved, since

$$\mathbf{x}_f - F^k \mathbf{x}_0 \in \text{Im} \mathcal{R}_k = \text{Im} \mathcal{R}_k \mathcal{R}_k^T,$$

and one can easily verify that all the solutions η to (5.45) yield, by means of (5.46), the same solution to (5.43). While (5.45) can have more than one solution, they all correspond, via (5.46), to a unique solution $\bar{\mathbf{u}}$ to (5.43). In fact, if both η_1 and η_2 are solutions to (5.45), then $\eta_1 - \eta_2 \in \ker(\mathcal{R}_k \mathcal{R}_k^T) = \ker \mathcal{R}_k^T$, and therefore $\mathcal{R}_k^T \eta_1 = \mathcal{R}_k^T \eta_2$.

If we consider as norm of the vector $\mathbf{u} = [\mathbf{u}^T(k-1) \quad \mathbf{u}^T(k-2) \quad \dots \quad \mathbf{u}^T(0)]^T \in \mathbb{R}^{mk}$ the norm induced by the standard internal product in the input space

$$\|\mathbf{u}\|_2 = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \left(\sum_{\sigma=0}^{k-1} \mathbf{u}^T(\sigma) \mathbf{u}(\sigma) \right)^{1/2},$$

the specific solution $\bar{\mathbf{u}}$, provided by (5.45) and (5.46), exhibits certain optimality feature if compared with the generic element of the set of solutions to (5.43), as the following result establishes.

Proposition 5.4.1 *The solution $\bar{\mathbf{u}}$, obtained through (5.46), has minimal norm among all the solutions to (5.43).*

PROOF The solution $\bar{\mathbf{u}} = \mathcal{R}_k^T \eta$ belongs to $\text{Im} \mathcal{R}_k^T = (\ker \mathcal{R}_k)^\perp$, and any other solution can be expressed as $\bar{\mathbf{u}} + \mathbf{v}$, where $\mathbf{v}$, solution to (5.44), is a generic element of $\ker \mathcal{R}_k$. Since $\bar{\mathbf{u}}$ and $\mathbf{v}$ are orthogonal, it holds true $\|\bar{\mathbf{u}} + \mathbf{v}\|^2 = \|\bar{\mathbf{u}}\|^2 + \|\mathbf{v}\|^2$, which shows that $\bar{\mathbf{u}}$ is the minimum norm solution. ■

If the system is reachable in k steps, the matrix $\mathcal{R}_k$ has rank n , and the $n \times n$ matrix $\mathcal{W}_k = \mathcal{R}_k \mathcal{R}_k^T$ is invertible. Then, the solution to (5.45) and (5.46) can be directly expressed as

$$\bar{\mathbf{u}} = \mathcal{R}_k^T \eta = \mathcal{R}_k^T \mathcal{W}_k^{-1} (\mathbf{x}_f - F^k \mathbf{x}_0). \quad (5.47)$$

Example 5.4.1 Consider the problem of heating steel bars passing through an oven partitioned into five areas. The reference temperature is the environment's temperature. Assume that the heating cost in each area is proportional to the square of the temperature in the area.

By referring to the i -th area of the oven, with $i = 0, 1, \dots, 4$, let

- θ_i be the temperature of the bar when it enters the area, which coincides with the temperature at the exit of the $(i-1)$ -th area;
- u_i be the temperature of the oven in the i -th area;
- the relation between the input temperature θ_i and output temperature θ_{i+1} of the bar, and the oven temperature u_i be

$$\theta_{i+1} = \theta_i + \frac{u_i - \theta_i}{2} \quad i = 0, 1, \dots, 4. \quad (5.48)$$

We want to determine the sequence $u_0, u_1, \dots, u_4$ of temperature in the areas that heats the bar from environmental temperature $\theta_0 = 0^\circ\text{C}$ to to temperature $\theta_5 = 1000^\circ\text{C}$, at the same time minimising the heating cost $\sum_{i=0}^4 u_i^2$. First of all, rewrite equations (5.48) in the form of a reachable linear discrete-time system

$$\theta_{i+1} = \frac{1}{2} \theta_i + \frac{1}{2} u_i, \quad (5.49)$$

with $F = 1/2$ and $g = 1/2$.

In such a system we have

$$\mathcal{R}_5 = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} \end{bmatrix}$$

and, applying (5.47), we obtain the temperature sequence

$$\begin{bmatrix} u_4 \\ u_3 \\ u_2 \\ u_1 \\ u_0 \end{bmatrix} = \mathcal{R}_5^T \left(\mathcal{R}_5 \mathcal{R}_5^T \right)^{-1} 1000 = \begin{bmatrix} 1/2 \\ 1/4 \\ 1/8 \\ 1/16 \\ 1/32 \end{bmatrix} \left(\frac{1 - (1/32)^2}{3} \right)^{-1} 1000.$$

If the discrete-time system $\Sigma = (F, G, H)$ is reachable in k steps and $\bar{\mathbf{x}}$ is the state to which one wants to steer the system at time t starting from time 0 and state $\mathbf{0}$, equation (5.47) yields the input $\bar{\mathbf{u}} = \mathcal{R}_k^T \mathcal{W}_k^{-1} \bar{\mathbf{x}}$, whose norm $\|\cdot\|_2$ is minimum among all those that achieve such a goal. Interpreting $\|\mathbf{u}\|_2^2 = \sum_{\sigma=0}^{k-1} \mathbf{u}^T(\sigma) \mathbf{u}(\sigma)$ as the control energy¹ of the input $\mathbf{u}$ over the interval $[0, k-1]$, the minimum energy necessary to reach $\bar{\mathbf{x}}$ at time k is thus

$$\|\bar{\mathbf{u}}\|_2^2 = \bar{\mathbf{u}}^T \bar{\mathbf{u}} = \bar{\mathbf{x}}^T \mathcal{W}_k^{-1} \mathcal{R}_k \mathcal{R}_k^T \mathcal{W}_k^{-1} \bar{\mathbf{x}} = \bar{\mathbf{x}}^T \mathcal{W}_k^{-1} \bar{\mathbf{x}}. \quad (5.50)$$

Introduce² an orthogonal matrix $M = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$ whose columns are an orthonormal basis of eigenvectors for the positive definite Gramian matrix $\mathcal{W}_k$. It follows that such a matrix factorises as

$$\mathcal{W}_k = M \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2\} M^T, \quad (5.51)$$

whose singular values σ_i are the roots of the (strictly positive) eigenvalues of $\mathcal{W}_k$, and which can be assumed to be ordered in decreasing order, possibly by permuting the columns of M . The minimum energy input among all those that steer the system into the state $\mathbf{x}(k) = \bar{\mathbf{x}}$ can be then rewritten as

$$\bar{\mathbf{u}}(\sigma) = G^T (F^T)^{k-1-\sigma} M \text{diag}\{\sigma_1^{-2}, \sigma_2^{-2}, \dots, \sigma_n^{-2}\} M^T \bar{\mathbf{x}}$$

and its energy is

$$\|\bar{\mathbf{u}}\|_2^2 = \bar{\mathbf{x}}^T M \text{diag}\{\sigma_1^{-2}, \sigma_2^{-2}, \dots, \sigma_n^{-2}\} M^T \bar{\mathbf{x}}. \quad (5.52)$$

Equation (5.52) suggests to reconsider the problem by describing the state space with respect to a different basis, obtained by using M as matrix of change of basis. In the algebraically equivalent system

$$\hat{\Sigma} = (\hat{F}, \hat{G}, \hat{H}) := (M^T F M, M^T G, H T)$$

each state $\hat{\mathbf{x}} = M^T \mathbf{x}$ of $\hat{\Sigma}$ has the same norm as the corresponding $\mathbf{x}$ in Σ , because M is an orthogonal matrix. However, (5.51) and (5.52) become simpler. In fact

- the reachability matrix $\hat{\mathcal{R}}_k = M^T \mathcal{R}_k$ of $\hat{\Sigma}$ yields a diagonal Gramian matrix

$$\hat{\mathcal{W}}_k = M^T \mathcal{R}_k \mathcal{R}_k^T M = \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2\}. \quad (5.53)$$

- reaching the state $\hat{\mathbf{x}}$ in $\hat{\Sigma}$ requires minimum energy

$$\|\bar{\mathbf{u}}\|_2^2 = \hat{\mathbf{x}}^T \text{diag}\{\sigma_1^{-2}, \dots, \sigma_n^{-2}\} \hat{\mathbf{x}}, \quad (5.54)$$

¹Compare the analysis provided in Example 5.5.2.

²Following the procedure proposed in Examples A.12.4 and A.12.6.

which corresponds to the optimal input $\bar{\mathbf{u}} = \hat{\mathcal{R}}_k \text{diag}\{\sigma_1^{-2}, \dots, \sigma_n^{-2}\} \hat{\mathbf{x}}$.

Among all states with unitary norm, the states $\mathbf{e}_1$ and $\mathbf{e}_n$ of $\hat{\Sigma}$ are those that require, respectively, the minimum and maximum control energy to be reached. Intuitively, therefore, the smallest singular values of the Gramian $\hat{\mathcal{W}}_k$ correspond to states that require the largest energy to be reached.

If we employ inputs $\mathbf{u}$ with energy $\|\mathbf{u}\|_2^2 \leq 1$, the states that the system $\hat{\Sigma}$ can reach in k steps are all and only those belonging to the ellipsoid whose hypersurface is described by

$$E^R = \{\hat{\mathbf{x}} : \hat{\mathbf{x}}^T \text{diag}\{\sigma_1^{-2}, \sigma_2^{-2}, \dots, \sigma_n^{-2}\} \hat{\mathbf{x}} = 1\} = \{\hat{\mathbf{x}} : \|\text{diag}\{\sigma_1^{-1}, \sigma_2^{-1}, \dots, \sigma_n^{-1}\} \hat{\mathbf{x}}\|_2 = 1\}.$$

When applying inputs with unitary energy over the interval $[0, k-1]$, the singular value σ_i corresponds to the maximum distance from the origin that the system $\hat{\Sigma}$ can reach in k steps along the direction of the canonical vector $\mathbf{e}_i$, and in order to reach the point $\sigma_i \mathbf{e}_i$ of the hypersurface E^R it is necessary to use one of the (unitary energy) optimal inputs $\bar{\mathbf{u}}$. From the inequalities $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$ it also clearly follows that $\sigma_1 \mathbf{e}_1$ is the maximum norm state that can be reached in k steps, while along the direction of $\mathbf{e}_n$ the norm of a reachable state cannot exceed σ_n .

- EXERCISE 5.4.1 Consider the system Σ and the matrix M in (5.51). Verify that if one can only use unitary norm inputs, then the maximum norm state that can be reached in k steps is $\mathbf{x}(k) = \sigma_1 \mathbf{v}_1$.
- EXERCISE 5.4.2 [NORMALISATION WITH RESPECT TO THE INPUT] Consider the system Σ as in the previous exercise. Set $S := \text{diag}\{\sigma_1, \sigma_2, \dots, \sigma_n\}$ and the matrix of change of basis $T := SM$. In the system that has been “normalised with respect to the input” $\Sigma^{IN} = (F^{IN}, G^{IN}, H^{IN}) := (T^{-1}FT, T^{-1}G, HT)$
 - (i) the k -steps reachability matrix is $\mathcal{R}_k^{IN} = (MS)^{-1} \mathcal{R}_k = S^{-1} M^T \mathcal{R}_k$ and the corresponding Gramian matrix is $\mathcal{W}_k^{IN} = I_n$;
 - (ii) k -long unitary norm inputs allow to reach, at time k , all and only the states whose norm is smaller than or equal to 1.

Notice that the matrix T is not orthonormal, and thus it does not preserve the norm of the state vectors.

We will now sketch the solution of the analogous problems for a continuous-time system

$$\dot{\mathbf{x}}(t) = F\mathbf{x}(t) + G\mathbf{u}(t).$$

Given a time horizon $[0, t]$, an initial state $\mathbf{x}(0) = \mathbf{x}_0$, and a final state $\mathbf{x}(t) = \mathbf{x}_f$, the goal is to study under which conditions there exists an input $\mathbf{u}(\cdot)$ inducing, over the interval $[0, t]$, a state evolution from $\mathbf{x}_0$ to $\mathbf{x}_f$, and, if it exists, to determine its structure.

Since the state at time t is the sum of the unforced evolution and of the forced evolution, and the states that can be reached at time t in forced evolution are the elements of X^R , there exists an input $\mathbf{u}(\cdot)$ that solves such a problem if and only if

$$\mathbf{x}_f - e^{Ft} \mathbf{x}_0 \in X^R. \quad (5.55)$$

Then, the states that can be reached at time t when starting from the initial state $\mathbf{x}_0$ are all and only the elements of the set $e^{Ft} \mathbf{x}_0 + X^R$. In case $\mathbf{x}_f$ belongs to such a set, we can determine the control input by solving, for $\mathbf{u}(\cdot) \in \mathcal{U}_{[0,t]}$, the equation

$$\mathbf{x}_f - e^{Ft} \mathbf{x}_0 = \int_0^t e^{F(t-\sigma)} G \mathbf{u}(\sigma) d\sigma = R_t \mathbf{u}(\cdot). \quad (5.56)$$

To solve such an equation we proceed analogously to the discrete-time case. In fact, the solutions are still given by the sum of a specific solution of (5.56) and of an arbitrary solution of the associated homogeneous system:

$$\mathbf{0} = R_t \mathbf{u}(\cdot). \quad (5.57)$$

A specific solution can be obtained by employing the adjoint operator and the Gramian matrix $\mathcal{W}_t = \int_0^t e^{F\tau} G G^T e^{F^T \tau} d\tau$ as follows

- solve, in the unknown $\eta \in \mathbb{R}^n$, the equation

$$R_t R_t^* \eta = \mathcal{W}_t \eta = \mathbf{x}_f - e^{Ft} \mathbf{x}_0 \quad (5.58)$$

- set

$$\bar{\mathbf{u}}(\cdot) = R_t^* \eta \quad (5.59)$$

to obtain the input

$$\bar{\mathbf{u}}(\sigma) = G^T e^{F^T(t-\sigma)} \eta, \quad \sigma \in [0, t]. \quad (5.60)$$

One can then verify that (5.58) and (5.59) determine a unique solution to (5.56), which is in particular the minimum norm solution once we set

$$\|\mathbf{u}(\cdot)\|_2^2 = \langle \mathbf{u}(\cdot), \mathbf{u}(\cdot) \rangle = \int_0^t \mathbf{u}^T(\sigma) \mathbf{u}(\sigma) d\sigma. \quad (5.61)$$

As for the discrete-time system, the quantity (5.61) will be called the “energy” of the input.

In particular, when the system (F, G) is reachable, then $\mathcal{W}_t$ is invertible, and (5.60) becomes

$$\bar{\mathbf{u}}(\sigma) = G^T e^{F^T(t-\sigma)} \mathcal{W}_t^{-1} (\mathbf{x}_f - e^{Ft} \mathbf{x}_0) \quad \sigma \in [0, t]. \quad (5.62)$$

Consequently, the minimum energy input that yields in forced evolution the state $\bar{\mathbf{x}}$ at time t is

$$\bar{\mathbf{u}}(\sigma) = G^T e^{F^T(t-\sigma)} \mathcal{W}_t^{-1} \bar{\mathbf{x}}, \quad \sigma \in [0, t],$$

and the input energy is

$$\|\bar{\mathbf{u}}\|_2^2 = \bar{\mathbf{x}}^T \mathcal{W}_t^{-1} \bar{\mathbf{x}}.$$

As in the discrete-time case, it is sometimes convenient to factorise the Gramian as in (5.51)

$$\mathcal{W}_t = M \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2\} M^T,$$

and employ the orthogonal matrix M to perform a change of basis in the state space.

In the new system $\hat{\Sigma} = (M^T F M, M^T G, H T) = (\hat{F}, \hat{G}, \hat{H})$ the reachability Gramian is the positive definite diagonal matrix

$$\hat{\mathcal{W}}_t = \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2\}$$

whose singular values σ_i depend on the duration of the control interval. Furthermore, the minimum energy input that allows to control the system state to $\hat{\mathbf{x}}$ at time t is

$$\bar{\mathbf{u}}(\sigma) = \hat{G}^T e^{\hat{F}^T(t-\sigma)} \text{diag}\{\sigma_1^{-2}, \sigma_2^{-2}, \dots, \sigma_n^{-2}\} \hat{\mathbf{x}} \quad \sigma \in [0, t],$$

with energy

$$\|\bar{\mathbf{u}}\|_2^2 = \hat{\mathbf{x}}^T \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2\} \hat{\mathbf{x}}.$$

Example 5.4.2 Consider the electrical circuit in Figure 5.5.1. The goal is to compute which voltage $v(\cdot)$ has to be applied over the interval $[0, 1]$ in order to steer the capacitor voltage from $v(0) = 0$ to $v(1) = 6$ Volt, while minimising the index $\int_0^1 v^2(\sigma) d\sigma$. Notice that such an index does not represent the energy supplied by the generator to the RC network over the horizon $[0, 1]$, which, instead, is given by

$$J = \int_0^1 v(\sigma) i(\sigma) d\sigma = \int_0^1 [v^2(\sigma) - v(\sigma) v_C(\sigma)] d\sigma.$$

Determining the input that minimises J is a typical optimal control problem, which will be the main focus of one of the next chapters.

The network equation

$$\dot{v}_C(t) = -v_C(t) + v(t)$$

can be interpreted as a linear state-space model with state variable v_C , input v , and matrices $F = -1$ and $G = 1$. The reachability Gramian over the interval $[0, 1]$ is

$$\mathcal{W}_1 = \int_0^1 e^{F\sigma} G G^T e^{F^T\sigma} d\sigma = \int_0^1 e^{-2\sigma} d\sigma = \frac{1 - e^{-2}}{2}.$$

By employing (5.62), we get

$$v(\sigma) = G^T e^{F^T(1-\sigma)} \mathcal{W}_1^{-1} v(1) = \frac{12e^\sigma}{e - e^{-1}}, \quad \sigma \in [0, 1].$$

- **EXERCISE 5.4.3** Consider the circuit in the previous Example, and assume it to be supplied by a current generator $j(\cdot)$. Determine the input function $j(\cdot)$ that steers the capacitor voltage from the initial value $v_C(0) = v_0$ to the final value $v_C(1) = v_f$, meanwhile minimising the dissipated energy $\int_0^1 R i^2(\sigma) d\sigma$.
- **EXERCISE 5.4.4** Prove that (5.60) has minimum norm among all the inputs that steer the continuous-time system from the state $\mathbf{x}_0$ to the state $\mathbf{x}_f$ over $[0, t]$.
 $\sharp$ Hint: any other such input is the sum of $\bar{\mathbf{u}}(\cdot) \in \text{Im} R_t^*$ and $\mathbf{v}(\cdot) \in \ker R_t = (\text{Im} R_t^*)^\perp$.

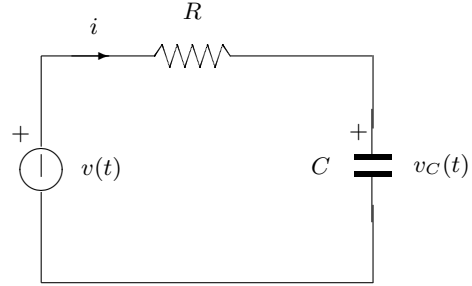
Remark 1 If the final state $\mathbf{x}_f$ is zero, the control law (5.60) can be expressed as a feedback from the state, namely, in such a way that the instantaneous input value $\bar{\mathbf{u}}(\sigma)$ linearly depends on the value $\mathbf{x}(\sigma)$ of the state at the same time.

In order to establish such a property, we observe first of all that, by (5.58), the initial state is related to the vector η by

$$\mathbf{x}_0 = -e^{-Ft} \mathcal{W}_t \eta. \quad (5.63)$$

By replacing in the state evolution both (5.63) and the control law (5.60) we have

$$\begin{aligned} \mathbf{x}(\sigma) &= e^{F\sigma} \mathbf{x}_0 + \int_0^\sigma e^{F(\sigma-\xi)} G \bar{\mathbf{u}}(\xi) d\xi = -e^{F(\sigma-t)} \mathcal{W}_t \eta + \int_0^\sigma e^{F(\sigma-\xi)} G (G^T e^{F^T(t-\xi)} \eta) d\xi \\ &= -\left(e^{-F(t-\sigma)} \mathcal{W}_t - e^{-F(t-\sigma)} \int_0^\sigma e^{F(t-\xi)} G G^T e^{F^T(t-\xi)} d\xi \right) \eta \end{aligned}$$



$$R = 1 \, \Omega, \, C = 1\text{F}$$

Figure 5.4.1

$$\begin{aligned}
&= -e^{-F(t-\sigma)} \left(\mathcal{W}_t - \int_0^t e^{F(t-\xi)} G G^T e^{F^T(t-\xi)} d\xi + \int_\sigma^t e^{F(t-\xi)} G G^T e^{F^T(t-\xi)} d\xi \right) \eta \\
&= -e^{-F(t-\sigma)} \mathcal{W}_{\sigma,t} \eta \quad \sigma \in [0, t],
\end{aligned} \tag{5.64}$$

where $\mathcal{W}_{\sigma,t} := \int_\sigma^t e^{F(t-\xi)} G G^T e^{F^T(t-\xi)} d\xi$ is the reachability Gramian for the interval $[\sigma, t]$. If the system is reachable, then $\mathcal{W}_{\sigma,t}$ is invertible for any $\sigma < t$. Then we obtain, by (5.64),

$$\eta = -\mathcal{W}_{\sigma,t}^{-1} e^{F(t-\sigma)} \mathbf{x}(\sigma),$$

which, plugged into (5.60), yields the following form for the control input $\bar{\mathbf{u}}(\sigma)$

$$\bar{\mathbf{u}}(\sigma) = [-G^T e^{F^T(t-\sigma)} \mathcal{W}_{\sigma,t}^{-1} e^{F(t-\sigma)}] \mathbf{x}(\sigma) = K(\sigma) \mathbf{x}(\sigma), \quad \sigma \in [0, t], \tag{5.65}$$

which is, as required, a linear feedback from the state at the same time.

- **EXERCISE 5.4.5*** Consider the discrete-time system $\mathbf{x}(t+1) = F\mathbf{x}(t) + G\mathbf{u}(t)$, and assume it to be reachable in k steps. Show that it is possible to express the input that steers the system from an arbitrary initial state $\mathbf{x}(0)$ into the final state $\mathbf{x}(k) = \mathbf{0}$ as a linear (and, in general, time-varying) function of the state, namely as $\mathbf{u}(\sigma) = K(\sigma)\mathbf{x}(\sigma)$, $\sigma = 0, 1, \dots, k-1$.

‡ *Solution.* Determining an input sequence $\bar{\mathbf{u}}_{[0,k-1]}$ that steers the system to $\mathbf{x}(k) = \mathbf{0}$ involves solving in the unknown η_k the equation $-F^k \mathbf{x}(0) = \mathcal{R}_k \mathcal{R}_k^T \eta_k = \mathcal{W}_k \eta_k$, and putting $\bar{\mathbf{u}}_{[0,k-1]} = \mathcal{R}_k^T \eta_k = -\mathcal{R}_k^T \mathcal{W}_k^{-1} F^k \mathbf{x}(0)$. Notice that, in particular, the last block of $\bar{\mathbf{u}}_{[0,k-1]}$,

$$\mathbf{u}(0) = -G^T (F^T)^{k-1} \mathcal{W}_k^{-1} F^k \mathbf{x}(0) = K(0) \mathbf{x}(0),$$

steers the system to a state $\mathbf{x}(1) \in X_{k-1}^C$.

As a consequence, there also exists an input sequence $\bar{\mathbf{u}}_{[1,k-1]}$ that steers the system from $\mathbf{x}(1)$ to $\mathbf{x}(k) = \mathbf{0}$. In order to obtain it, one can solve in the unknown η_{k-1} the equation $-F^{k-1} \mathbf{x}(1) = \mathcal{R}_{k-1} \mathcal{R}_{k-1}^T \eta_{k-1} = \mathcal{W}_{k-1} \eta_{k-1}$. This can be done, for example, by employing the Moore-Penrose pseudoinverse of the Gramian (see A.13), and putting $\bar{\mathbf{u}}_{[1,k-1]} = \mathcal{R}_{k-1}^T \eta_{k-1} = -\mathcal{R}_{k-1}^T \mathcal{W}_{k-1}^\dagger F^{k-1} \mathbf{x}(1)$. Notice now that, again, the last block of $\bar{\mathbf{u}}_{[1,k-1]}$,

$$\mathbf{u}(1) = -G^T (F^T)^{k-2} \mathcal{W}_{k-1}^\dagger F^{k-1} \mathbf{x}(1) = K(1) \mathbf{x}(1),$$

steers the system to the state $\mathbf{x}(2) \in X_{k-2}^C$. By induction, then, it is possible to show that the required input is given by $\mathbf{u}(t) = -G^T (F^T)^{k-t-1} \mathcal{W}_{k-t}^\dagger F^{k-t} \mathbf{x}(t) = K(t) \mathbf{x}(t)$, $t = 0, 1, \dots, k-1$.

5.5 Control and Gramian matrices over infinite horizon

In a discrete-time system $\Sigma = (F, G, H)$, if a state is reachable from $\mathbf{0}$ in k steps with input $\mathbf{u}(0), \dots, \mathbf{u}(k-1)$, then it is reachable in $k+1$ steps with input $\mathbf{v}(0) = \mathbf{0}, \mathbf{v}(1) = \mathbf{u}(0), \dots, \mathbf{v}(k) = \mathbf{u}(k-1)$, with the same norm. Consequently, the minimum energy of an input that allows to reach, in forced evolution, a reachable state $\mathbf{x}$ is a non-increasing function of the duration of the control interval, and the minimum energy that is necessary to reach $\mathbf{x}$ tends towards a finite limit when the duration of the control interval tends to infinite. The next result is instrumental in determining such limit value.

Lemma 5.5.1 [LYAPUNOV EQUATION AND REACHABILITY] *Given a discrete-time system $\Sigma = (F, G, -)$ of dimension n , any two of the following statements imply the third:*

- Σ is reachable;

ii) the Lyapunov equation

$$FXF^T - X = -GG^T \quad (5.66)$$

admits a positive definite solution;

iii) $|\Lambda(F)| < 1$, namely, all eigenvalues of F have modulus smaller than 1.

PROOF (i) and (ii) $\Rightarrow$ (iii) Let P be a positive definite solution to (5.66), and, for the autonomous linear system

$$\mathbf{z}(t+1) = F^T \mathbf{z}(t) \quad (5.67)$$

consider the Lyapunov function $V(\mathbf{z}) = \mathbf{z}^T P \mathbf{z}$. We obtain

$$\Delta V(\mathbf{z}) = V(F^T \mathbf{z}) - V(\mathbf{z}) = \mathbf{z}^T F P F^T \mathbf{z} - \mathbf{z}^T P \mathbf{z} = -\mathbf{z}^T G G^T \mathbf{z},$$

wherefore ΔV is negative semi-definite and the origin is at least a stable equilibrium for (5.67). Assume, by contradiction, that the set

$$\mathcal{N} := \{\mathbf{z} : \Delta V(\mathbf{z}) = 0\} = \{\mathbf{z} : \mathbf{z}^T G = \mathbf{0}^T\}$$

includes a perturbed state $\mathbf{z}_0 \neq \mathbf{0}$, from which it stems a trajectory $\mathbf{z}_0, F^T \mathbf{z}_0, (F^T)^2 \mathbf{z}_0, \dots$ entirely contained in $\mathcal{N}$. Then

$$\mathbf{0}^T = \mathbf{z}_0^T G = \mathbf{z}_0^T F G = \mathbf{z}_0^T F^2 G = \dots$$

and therefore

$$\mathbf{z}_0^T [G \quad FG \quad F^2 G \quad \dots \quad F^{n-1} G] = \mathbf{0}$$

which, however, contradicts the reachability property of Σ . By Krasowskii's stability criterion, one concludes that the origin is an asymptotically stable equilibrium for (5.67), and therefore

$$|\Lambda(F^T)| = |\Lambda(F)| < 1.$$

(ii) and (iii) $\Rightarrow$ (i) If $|\Lambda(F)| < 1$, we can extend in a straightforward way the argument presented in Section 3.5 in order to show that

- the series $\sum_{i=0}^{\infty} F^i G G^T (F^T)^i$ converges,
- its sum, which we denote by $\mathcal{W}_{\infty}$, solves the Lyapunov equation (5.66),
- the solution to the equation is unique.

Therefore, $\mathcal{W}_{\infty}$ is a positive definite matrix.

If the pair (F, G) were not reachable, it would exist a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ satisfying

$$\mathbf{x}^T [G \quad FG \quad \dots \quad F^{n-1} G] = \mathbf{0}.$$

By Cayley-Hamilton's theorem, this implies that $\mathbf{x}^T F^i G = \mathbf{0}$, $\forall i \geq 0$, which in turn yields $\mathbf{x}^T \mathcal{W}_{\infty} \mathbf{x} = 0$, contradicting the positive definiteness of $\mathcal{W}_{\infty}$.

(iii) and (i) $\Rightarrow$ (ii) As in the previous part of the proof, from $|\Lambda(F)| < 1$ it follows that the series $\sum_{i=0}^{\infty} F^i G G^T (F^T)^i$ converges, and its limit value, $\mathcal{W}_{\infty}$, solves equation (5.66). Furthermore, reachability of (F, G) implies that the n -steps Gramian

$$\mathcal{W}_n = \mathcal{R} \mathcal{R}^T = \sum_{i=0}^{n-1} F^i G G^T (F^T)^i$$

is not singular, and therefore positive definite. Therefore, the solution to the Lyapunov equation (5.66), $\mathcal{W}_\infty = \sum_{i=0}^{\infty} F^i G G^T (F^T)^i$, is positive definite. ■

We can now determine what is the minimum energy that is necessary to steer the system from the state $\mathbf{0}$ to an arbitrary state $\bar{\mathbf{x}} \in \mathbb{R}^n$ in a system (F, G, H) that is reachable and internally stable (i.e., $|\Lambda(F)| < 1$) as the duration of the control interval diverges, and we can compute the input that implements such an action.

Proposition 5.5.2 [REACHABILITY GRAMIAN OVER INFINITE HORIZON] *If the discrete-time system $\Sigma = (F, G, H)$ is reachable and internally stable, for any $\bar{\mathbf{x}} \in \mathbb{R}^n$*

i) *the input $\bar{\mathbf{u}}(\cdot)$ defined over the horizon $(-\infty, 0)$ by*

$$\bar{\mathbf{u}}(-\sigma) = G^T (F^T)^{\sigma-1} \mathcal{W}_\infty^{-1} \bar{\mathbf{x}}, \quad \forall \sigma > 0,$$

steers the system from the state $\mathbf{x}(-\infty) = \mathbf{0}$ to the state $\mathbf{x}(0) = \bar{\mathbf{x}}$, minimising the control energy $\|\bar{\mathbf{u}}\|_2^2 = \sum_{\sigma=1}^{\infty} \bar{\mathbf{u}}^T(-\sigma) \bar{\mathbf{u}}(-\sigma)$;

ii) *the energy of $\bar{\mathbf{u}}(\cdot)$ is given by*

$$\|\bar{\mathbf{u}}\|_2^2 = \bar{\mathbf{x}}^T \mathcal{W}_\infty^{-1} \bar{\mathbf{x}},$$

where the matrix $\mathcal{W}_\infty = \sum_{i=0}^{\infty} F^i G G^T (F^T)^i$, called the reachability Gramian of Σ over infinite horizon, is the positive definite solution to the Lyapunov equation (5.66).

PROOF For any value of $k \geq n$, the input $\mathbf{u}^{(k)}(\cdot)$ with minimum energy that steers the system from $\mathbf{x}(-k) = \mathbf{0}$ into $\mathbf{x}(0) = \bar{\mathbf{x}}$ is given by (5.47)

$$\begin{bmatrix} \mathbf{u}^{(k)}(-1) \\ \mathbf{u}^{(k)}(-2) \\ \vdots \\ \mathbf{u}^{(k)}(-k) \end{bmatrix} = \mathcal{R}_k^T \mathcal{W}_k^{-1} \bar{\mathbf{x}} = \begin{bmatrix} G^T \\ G^T F^T \\ \vdots \\ G^T (F^T)^{k-1} \end{bmatrix} \mathcal{W}_k^{-1} \bar{\mathbf{x}}$$

and therefore by $\mathbf{u}^{(k)}(-\sigma) = G^T (F^T)^{\sigma-1} \mathcal{W}_k^{-1} \bar{\mathbf{x}}$, $\sigma = 1, 2, \dots, k$.

By the previous lemma, as k diverges the positive definite Gramian

$$\mathcal{W}_k = \sum_{i=0}^{k-1} F^i G G^T (F^T)^i$$

tends to the positive definite (and thus invertible) matrix

$$\mathcal{W}_\infty = \sum_{i=0}^{\infty} F^i G G^T (F^T)^i,$$

that solves the Lyapunov equation (5.66), and for any $\sigma > 0$ the input $\mathbf{u}^{(k)}(-\sigma)$ has limit

$$\bar{\mathbf{u}}(-\sigma) = \lim_{k \rightarrow \infty} G^T (F^T)^{\sigma-1} \mathcal{W}_k^{-1} \bar{\mathbf{x}} = G^T (F^T)^{\sigma-1} \mathcal{W}_\infty^{-1} \bar{\mathbf{x}}. \quad (5.68)$$

Equation (5.68) defines an input $\bar{\mathbf{u}}(\cdot)$ with support in $(-\infty, 0)$ and energy

$$\|\bar{\mathbf{u}}\|_2^2 = \sum_{\sigma=1}^{\infty} \bar{\mathbf{x}}^T \mathcal{W}_{\infty}^{-1} F^{\sigma-1} G G^T (F^T)^{\sigma-1} \mathcal{W}_{\infty}^{-1} \bar{\mathbf{x}} = \bar{\mathbf{x}}^T \mathcal{W}_{\infty}^{-1} \bar{\mathbf{x}} = \lim_{k \rightarrow \infty} \bar{\mathbf{x}}^T \mathcal{W}_k^{-1} \bar{\mathbf{x}}.$$

Implementing such an input³ produces the final state

$$\mathbf{x}(0) = \sum_{\sigma=1}^{\infty} F^{\sigma-1} G \bar{\mathbf{u}}(-\sigma) = \sum_{\sigma=1}^{\infty} F^{\sigma-1} G G^T (F^T)^{\sigma-1} \mathcal{W}_{\infty}^{-1} \bar{\mathbf{x}} = \bar{\mathbf{x}}. \quad \blacksquare$$

- EXERCISE 5.5.1 Under the same assumptions as in Proposition 5.5.2, if $\bar{\mathbf{u}}^{(i)}(\cdot)$, $i = 1, 2$ are two minimum energy inputs that steer the system into the state vectors $\mathbf{x}^{(i)}$, $i = 1, 2$, then
 - (i) $\langle \bar{\mathbf{u}}^{(1)}(\cdot), \bar{\mathbf{u}}^{(2)}(\cdot) \rangle := \sum_{\sigma=1}^{\infty} (\mathbf{u}^{(1)})^T(-\sigma) \mathbf{u}^{(2)}(-\sigma) = (\mathbf{x}^{(1)})^T \mathcal{W}_{\infty}^{-1} \mathbf{x}^{(2)}$;
 - (ii) if the basis of the state space is chosen in such a way that $\mathcal{W}_{\infty}$ is a diagonal matrix, the minimum energy inputs that correspond to the canonical states $\mathbf{e}_i$ are orthogonal.

Under suitable adaptations, the analysis of the properties of the reachability Gramian extends to the continuous-time case. The details are left as exercise to the reader.

- EXERCISE 5.5.2 Consider the continuous-time n -dimensional system $\Sigma = (F, G, -)$. Any two of the following statements imply the third:
 - i) Σ is reachable;
 - ii) the Lyapunov equation $FX + XF^T = -GG^T$ admits a positive definite solution;
 - iii) $\operatorname{Re} \Lambda(F) < 0$, namely all eigenvalues of F have negative real part.

‡ Hint: Think along the same lines as for Lemma 5.5.1, with the difference that the sums are to be replaced by integrals, and the reachability Gramian over infinite horizon is $\mathcal{W}_{\infty} = \int_0^{\infty} e^{F\tau} G G^T e^{F^T \tau} d\tau$.
- EXERCISE 5.5.3 If the continuous-time system $\Sigma = (F, G, H)$ is reachable and internally stable, then
 - i) the input $\bar{\mathbf{u}}(\cdot)$ that is defined over the interval $(-\infty, 0)$ by $\bar{\mathbf{u}}(-\sigma) = G^T e^{F^T \sigma} \mathcal{W}_{\infty}^{-1} \bar{\mathbf{x}}$, $\forall \sigma \geq 0$, steers the system from the state $\mathbf{x}(-\infty) = \mathbf{0}$ to the state $\mathbf{x}(0) = \bar{\mathbf{x}}$ minimising the control energy $\|\bar{\mathbf{u}}\|_2^2 = \int_{\sigma=0}^{\infty} \bar{\mathbf{u}}^T(-\sigma) \bar{\mathbf{u}}(-\sigma) d\sigma$;
 - ii) the energy of $\bar{\mathbf{u}}(\cdot)$ is given by $\|\bar{\mathbf{u}}\|_2^2 = \bar{\mathbf{x}}^T \mathcal{W}_{\infty}^{-1} \bar{\mathbf{x}}$.

5.6 Control of movement in a subspace

In this section we consider another interesting problem, namely the possibility to maintain for an arbitrary time the state of a system $\Sigma = (F, G, H)$ in a given subspace V of the state space, when the initial state is an arbitrary $\mathbf{x}_0 \in V$. Such a subspace, within which the state remains for suitable input sequences, is called a *controlled invariant* of Σ , or equivalently an (F, G) -invariant subspace.

Proposition 4.6.1 [STRUCTURE OF CONTROLLED INVARIANT] *A subspace V of the state space of the system $\Sigma = (F, G, H)$ is a controlled invariant if and only if*

$$FV \subseteq V + \operatorname{Im} G. \quad (5.69)$$

PROOF We first address the case of a discrete-time system Σ .

If (5.69) holds true, and the state $\mathbf{x}(t)$ belongs to V , it is possible to express the vector $F\mathbf{x}(t)$ as

$$F\mathbf{x}(t) = \mathbf{v}(t) - \mathbf{g}(t), \quad \exists \mathbf{v}(t) \in V, \quad \exists \mathbf{g}(t) \in \operatorname{Im} G.$$

³By extending the superposition principle to infinite addends.

This means that, for a suitable vector $\mathbf{u}(t) \in \mathbb{R}^m$, it holds true

$$F\mathbf{x}(t) + \mathbf{g}(t) = F\mathbf{x}(t) + G\mathbf{u}(t) = \mathbf{v}(t) \in V.$$

Consequently, applying $\mathbf{u}(t)$ yields a state $\mathbf{x}(t+1) \in V$. By induction, if $\mathbf{x}(0)$ belongs to V , one can determine an input sequence that keeps in V the entire trajectory.

Conversely, if for any $\mathbf{x}(0) \in V$ there exists $\mathbf{u}(0) \in \mathbb{R}^m$ such that $\mathbf{x}(1) = F\mathbf{x}(0) + G\mathbf{u}(0)$ belongs to V , the vectors of FV can all be expressed as the sum of a vector of V and one of $\text{Im}G$, which implies (5.69).

We consider now a continuous time system $\dot{\mathbf{x}}(t) = F\mathbf{x}(t) + G\mathbf{u}(t)$, and we assume that (5.69) hold true. In addition, set $\dim V = \nu$, $\dim(V + \text{Im}G) = \nu + \mu$. Introduce the following basis of the state space

$$(\mathbf{v}_1, \dots, \mathbf{v}_\nu, \mathbf{g}_1, \dots, \mathbf{g}_\mu, \mathbf{w}_1, \dots, \mathbf{w}_{n-(\nu+\mu)}) \quad (5.70)$$

whose first ν vectors are a basis for V and the subsequent μ vectors, $\mathbf{g}_1, \dots, \mathbf{g}_\mu$ are in $\text{Im}G$ and provide, along with the first ν , a basis for $V + \text{Im}G$.

With respect to such a basis in X , the representations of vectors in $V + \text{Im}G$ are n -uples whose last $n - (\nu + \mu)$ entries are zero, while those of vectors in V are n -uples whose last $n - \nu$ entries are zero. Furthermore, the system equations $\dot{\mathbf{x}}(t) = F\mathbf{x}(t) + G\mathbf{u}(t)$ become

$$\dot{\bar{\mathbf{x}}}(t) = \bar{F}\bar{\mathbf{x}}(t) + \bar{G}\mathbf{u}(t) = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} & \bar{F}_{13} \\ \bar{F}_{21} & \bar{F}_{22} & \bar{F}_{23} \\ 0 & \bar{F}_{32} & \bar{F}_{33} \end{bmatrix} \bar{\mathbf{x}}(t) + \begin{bmatrix} \bar{G}_1 \\ \bar{G}_2 \\ 0 \end{bmatrix} \mathbf{u}(t), \quad (5.71)$$

where the structure of F follows from the fact that, by assumption, any vector of V is transformed, by F , into a vector of $V + \text{Im}G$.

Let now $M \in \mathbb{R}^{m \times \mu}$ be a matrix of rank μ such that $[\mathbf{g}_1 \dots \mathbf{g}_\mu] = GM$, and control the system (5.71) by means of the static “pre- controller” M , namely, by means of the control law

$$\mathbf{u}(t) = M\mathbf{i}(t). \quad (5.72)$$

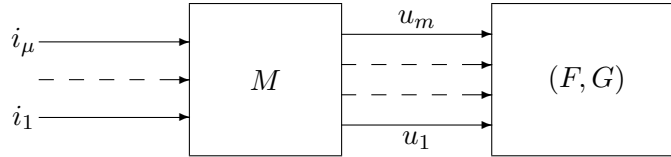


Figure 4.6.1

With respect to the basis (5.70) the columns of GM are represented by the vectors $\mathbf{e}_{\nu+1}, \dots, \mathbf{e}_{\nu+\mu}$ of the canonical basis of $\mathbb{R}^n$, and the equations of the closed-loop system become

$$\begin{bmatrix} \dot{\bar{\mathbf{x}}}_1(t) \\ \dot{\bar{\mathbf{x}}}_2(t) \\ \dot{\bar{\mathbf{x}}}_3(t) \end{bmatrix} = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} & \bar{F}_{13} \\ \bar{F}_{21} & \bar{F}_{22} & \bar{F}_{23} \\ 0 & \bar{F}_{32} & \bar{F}_{33} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{x}}_1(t) \\ \bar{\mathbf{x}}_2(t) \\ \bar{\mathbf{x}}_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ I_\mu \\ 0 \end{bmatrix} \mathbf{i}(t). \quad (5.73)$$

If we prove that V is a controlled invariant for (5.73), then the same clearly holds for (5.71). To this aim, set $\mathbf{i}(t) = -\bar{F}_{21}\bar{\mathbf{x}}_1(t) \in \mathbb{R}^\mu$, for all $t \geq 0$. The dynamics of the state is the same as the system in unforced evolution

$$\begin{bmatrix} \dot{\bar{\mathbf{x}}}_1(t) \\ \dot{\bar{\mathbf{x}}}_2(t) \\ \dot{\bar{\mathbf{x}}}_3(t) \end{bmatrix} = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} & \bar{F}_{13} \\ 0 & \bar{F}_{22} & \bar{F}_{23} \\ 0 & \bar{F}_{32} & \bar{F}_{33} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{x}}_1(t) \\ \bar{\mathbf{x}}_2(t) \\ \bar{\mathbf{x}}_3(t) \end{bmatrix} = \tilde{F}\bar{\mathbf{x}}(t), \quad (5.74)$$

in which the subspace V is $\tilde{F}$ -invariant. Therefore, if $\bar{\mathbf{x}}(0)$ belongs to V , the same necessarily holds true for $\bar{\mathbf{x}}(t)$, for any $t \geq 0$.

Conversely, assume that, for a given arbitrary state $\mathbf{x}(0) \in V$, there exists an input $\mathbf{u}(\cdot)$ such that $\mathbf{x}(t) \in V$ for any $t \geq 0$. Even if $\mathbf{x}(t)$ is not differentiable over a set of zero measure, the derivative $\dot{\mathbf{x}}(t)$, when it exists, necessarily belongs to V – to see this, it suffices to notice that the is always in V . Then $F\mathbf{x}(t) = \dot{\mathbf{x}}(t) - G\mathbf{u}(t) \in V + \text{Im}G$ holds true for almost all t , and by continuity of $\mathbf{x}(\cdot)$ it holds true $F\mathbf{x}(t) \in V + \text{Im}G$ for any $t \geq 0$. Then to show that V satisfies (5.69) it suffices to notice that the previous argument holds true for any $\mathbf{x}(0)$ in V . ■

- EXERCISE 4.6.1 Let (F, G) be a discrete-time system, and V be a subspace of the state space. Then
 - (i) X^R is a controlled invariant;
 - (ii) X^C is a controlled invariant;
 - (iii) if V is F -invariant and includes X^R , then any trajectory that starts in V is completely contained in V , for any control input;
 - (iv) if any trajectory that starts in V is contained in V , then V is F -invariant, and includes X^R .

Proposition 4.6.2 *If V_1 and V_2 are controlled invariants for the pair (F, G) , then the same holds true for $V_1 + V_2$.*

PROOF If the initial state is $\mathbf{x}_1(0) + \mathbf{x}_2(0)$, where $\mathbf{x}_i(0) \in V_i$, $i = 1, 2$, and if he inputs $\mathbf{u}_i(\cdot)$ keep the system in V_i starting from the initial states $\mathbf{x}_i(0)$, by the superposition principle it is sufficient to apply to the system the input $\mathbf{u}_1(\cdot) + \mathbf{u}_2(\cdot)$ to keep the trajectory in $V_1 + V_2$. ■

The intersection of two controlled invariants is not necessarily a controlled invariant, as the next example shows.

Example 4.6.1 Consider the discrete-time system

$$\mathbf{x}(t+1) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t) = F\mathbf{x}(t) + \mathbf{g}u(t).$$

The subspaces $V_1 = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ and $V_2 = \text{span}\{\mathbf{e}_1, \mathbf{e}_3\}$ are controlled invariants, since

$$V_1 + \text{Im}(\mathbf{g}) = V_2 + \text{Im}(\mathbf{g}) = \mathbb{R}^3.$$

However $V_1 \cap V_2 = \text{span } \mathbf{e}_1$ is not a controlled invariant, as

$$F\mathbf{e}_1 = \mathbf{e}_2 \notin \text{span}\{\mathbf{e}_1, \mathbf{g}\}.$$

If the subspace $V \subset X$ is not a controlled invariant, namely if it is not possible to keep in V the trajectory of the system starting from an arbitrary state in V , then

1. there exist nonetheless initial states in V starting from which it is possible to keep in V the entire system evolution - at the very least, it is possible to remain in the zero state by applying the zero input, and
2. the set of states of V that exhibit this property is a subspace of V^4 and it is obviously the largest invariant included in V . For this reason, it is called the *maximal (F, G) - controlled invariant contained in V* , or, when not ambiguous, the *maximal controlled invariant of V* , and denoted by V^* .

- EXERCISE 4.6.2 For any subspace V of X the maximal controlled invariant is unique.

5.7 Representation of non-reachable systems

Consider an n -dimensional – either discrete- or continuous-time – linear system $\Sigma = (F, G, H)$ which is not (completely) reachable, namely, for which $\dim X^R = \rho < n$. Consider a basis of the state space X , and assume that, among its n elements, the first ρ is a basis for X^R . Let T be the matrix of change of basis. Recalling that X^R is F -invariant and that $\text{Im} G \subseteq X^R$, such a change of basis yields a system $\bar{\Sigma} = (\bar{F}, \bar{G}, \bar{H})$, algebraically equivalent to Σ , for which

$$\begin{aligned}\bar{F} &= T^{-1}FT = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix}, & \bar{G} &= T^{-1}G = \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix} \\ \bar{H} &= HT = [\bar{H}_1 \quad \bar{H}_2],\end{aligned}\tag{5.75}$$

where $\bar{F}_{11} \in \mathbb{R}^{\rho \times \rho}$, $\bar{G}_1 \in \mathbb{R}^{\rho \times m}$, $\bar{H}_1 \in \mathbb{R}^{p \times \rho}$, and in which the pair $(\bar{F}_{11}, \bar{G}_1)$ is reachable. This can be established by noticing that the reachability matrix $\bar{\mathcal{R}}$ of $\bar{\Sigma}$

$$\bar{\mathcal{R}} = \begin{bmatrix} \bar{G}_1 & \bar{F}_{11}\bar{G}_1 & \dots & \bar{F}_{11}^{n-1}\bar{G}_1 \\ 0 & 0 & \dots & 0 \end{bmatrix} \} \rho \text{ rows}$$

is related to the reachability matrix $\mathcal{R}$ of Σ by means of $\bar{\mathcal{R}} = T^{-1}\mathcal{R}$, therefore it has the same rank ρ . It then follows that the first ρ rows of $\bar{\mathcal{R}}$ are linearly independent, and it is straightforward to show that the matrix

$$\bar{\mathcal{R}}^{(R)} = [\bar{G}_1 \quad \bar{F}_{11}\bar{G}_1 \quad \dots \quad \bar{F}_{11}^{\rho-1}\bar{G}_1]$$

has also rank ρ . The system $\bar{\Sigma}$ obtained in (5.75) is called the *standard reachability form* of Σ .

Remark 1 A system $\bar{\Sigma} = (\bar{F}, \bar{G}, \bar{H})$ is in if the following three conditions hold simultaneously

1. the matrix $\bar{F}$ is block triangular, and the block $\bar{F}_{21} \in \mathbb{R}^{(n-\rho) \times \rho}$ is zero;
2. the non-zero elements of the matrix $\bar{G}$ all belong to the submatrix $\bar{G}_1$ which is composed of the first ρ rows;
3. the pair $(\bar{F}_{11}, \bar{G}_1)$ is reachable.

⁴One can verify this statement in a way that is analogous to the proof of Proposition 4.6.2, with $V_1 = V_2 = V$.

For the discrete-time case⁵, set

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{\mathbf{x}}_1 \\ \bar{\mathbf{x}}_2 \end{bmatrix} \} \rho \text{ entries}$$

Then the equations of $\bar{\Sigma}$ can be rewritten as

$$\begin{aligned} \bar{\mathbf{x}}_1(t+1) &= \bar{F}_{11}\bar{\mathbf{x}}_1(t) + \bar{F}_{12}\bar{\mathbf{x}}_2(t) + \bar{G}_1\mathbf{u}(t) \\ \bar{\mathbf{x}}_2(t+1) &= \bar{F}_{22}\bar{\mathbf{x}}_2(t) \\ \mathbf{y}(t) &= \bar{H}_1\bar{\mathbf{x}}_1(t) + \bar{H}_2\bar{\mathbf{x}}_2(t) \end{aligned} \quad (5.76)$$

and such a system corresponds to the block diagram in Figure 4.7.1.

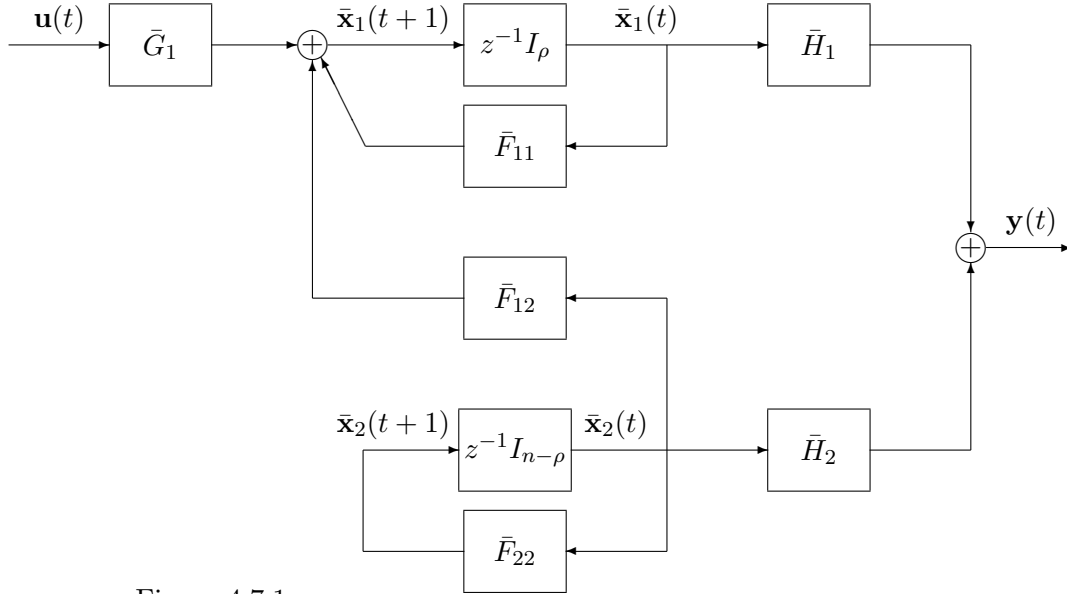


Figure 4.7.1

The subsystem $\bar{\Sigma}^{(R)} = (\bar{F}_{11}, \bar{G}_1, \bar{H}_1)$, whose dimension is ρ , is called the *reachable subsystem*, and completely describes the trajectory of $\bar{\Sigma}$ if the initial state belongs to the reachable subspace, namely when $\bar{\mathbf{x}}_2(0) = \mathbf{0}$, since in this case $\bar{\mathbf{x}}_2(t) = \mathbf{0}$ for any $t > 0$. Viceversa, the dynamics of the subsystem $\bar{\Sigma}^{(NR)} = (\bar{F}_{22}, \mathbf{0}, \bar{H}_2)$, called the *non-reachable subsystem*, only depends on the initial state $\bar{\mathbf{x}}_2(0)$, and cannot be influenced by the control input.

Remark 2 Assume $\bar{\mathbf{x}}_2(0) \neq \mathbf{0}$. Then the term $\bar{F}_{12}\bar{\mathbf{x}}_2(\cdot)$, which represents the effect of the dynamics of $\bar{\Sigma}^{(NR)}$ on $\bar{\Sigma}^{(R)}$, is not zero, but nonetheless does not influence the capability of the input $\mathbf{u}(\cdot)$ to steer the reachable subsystem $\bar{\Sigma}^{(R)}$ from an arbitrary initial state $\bar{\mathbf{x}}_{1i}$ to an arbitrary final state $\bar{\mathbf{x}}_{1f}$ – in discrete-time, in k steps, for any $k \geq \rho$.

To show this, denote by $\tilde{\mathbf{x}}_1$ the state that the signal $\bar{\mathbf{x}}_2(\cdot)$, interpreted as an input, yields in forced evolution at time k in the linear system

$$\bar{\mathbf{x}}_1(t+1) = \bar{F}_{11}\bar{\mathbf{x}}_1(t) + \bar{F}_{12}\bar{\mathbf{x}}_2(t).$$

By the superposition principle, one can simply choose an input $\mathbf{u}(\cdot)$ which, on its own (namely, if $\bar{\mathbf{x}}_2(\cdot) = \mathbf{0}$), steers the system $\bar{\Sigma}^{(R)}$ from the state $\bar{\mathbf{x}}_{1i}$, at time 0, to the state $\bar{\mathbf{x}}_{1f} - \tilde{\mathbf{x}}_1$, at time k . To do so, one can

⁵In continuous-time the procedure is totally analogous.

solve, in the unknown $\mathbf{u} \in \mathbb{R}^{m_k}$, the equation

$$\bar{\mathbf{x}}_{1f} - \bar{\mathbf{x}}_1 - \bar{F}_{11}^k \bar{\mathbf{x}}_{1i} = [\bar{G}_1 \quad \bar{F}_{11} \bar{G}_1 \quad \dots \quad \bar{F}_{11}^{k-1} \bar{G}_1] \begin{bmatrix} \mathbf{u}(k-1) \\ \mathbf{u}(k-2) \\ \vdots \\ \mathbf{u}(0) \end{bmatrix} = \mathcal{R}_k^{(R)} \mathbf{u},$$

thus obtaining $\mathbf{u} = \left(\mathcal{R}_k^{(R)}\right)^T \left[\mathcal{R}_k^{(R)} \left(\mathcal{R}_k^{(R)}\right)^T\right]^{-1} (\bar{\mathbf{x}}_{1f} - \bar{\mathbf{x}}_1 - \bar{F}_{11}^k \bar{\mathbf{x}}_{1i})$.

The procedure in continuous-time is analogous. Determining the input $\mathbf{u}(\cdot)$ for this class of systems is left to the reader as an exercise.

The transfer matrix $W_{\bar{\Sigma}}$ of the system $\bar{\Sigma}$, and thus also of Σ , coincides with the transfer matrix $W_{\bar{\Sigma}(R)}$ of the reachable subsystem $\bar{\Sigma}^{(R)}$. In fact, it holds true

$$\begin{aligned} W_{\bar{\Sigma}}(z) &= [\bar{H}_1 \quad \bar{H}_2] \begin{bmatrix} zI_\rho - \bar{F}_{11} & -\bar{F}_{12} \\ 0 & zI_{n-\rho} - \bar{F}_{22} \end{bmatrix}^{-1} \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix} \\ &= [\bar{H}_1 \quad \bar{H}_2] \begin{bmatrix} (zI_\rho - \bar{F}_{11})^{-1} & * \\ 0 & (zI_{n-\rho} - \bar{F}_{22})^{-1} \end{bmatrix} \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix} \\ &= \bar{H}_1 (zI_\rho - \bar{F}_{11})^{-1} \bar{G}_1 = W_{\bar{\Sigma}(R)} \end{aligned} \quad (5.77)$$

Example 4.7.1 Consider the system

$$\begin{aligned} \mathbf{x}(t+1) &= F\mathbf{x}(t) + G\mathbf{u}(t) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \mathbf{u}(t) \\ y(t) &= H\mathbf{x}(t) = [1 \quad 1 \quad 1 \quad 1] \mathbf{x}(t). \end{aligned} \quad (5.78)$$

The reachability matrix

$$\mathcal{R} = [G \quad FG \quad F^2G \quad F^3G] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

has rank 3, and thus $\dim X^R = 3$. In order to reduce the system to standard reachability form, one can choose a basis whose first three vectors generate the subspace X^R . This can be done by extracting from $\mathcal{R}$ any three linearly independent columns, and adding then a fourth column which is linearly independent of the first three. One possible such change of basis matrix is

$$T = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{\text{columns of } \mathcal{R}}.$$

Since $T = T^{-1}$, computing the standard reachability form is straightforward, and we have

$$T^{-1}FT = \bar{F} = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right] = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix},$$

$$\begin{aligned}
T^{-1}G = \bar{G} &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ \text{---} & \text{---} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix} \\
HT = \bar{H} &= [1 \quad 1 \quad 1 \quad | \quad 1] = [\bar{H}_1 \quad \bar{H}_2].
\end{aligned}$$

The transfer matrix is

$$\begin{aligned}
W(z) &= \bar{H}_1(zI - \bar{F}_{11})^{-1}\bar{G}_1 \\
&= [1 \quad 1 \quad 1] \begin{bmatrix} z-1 & 0 & 0 \\ 0 & z & -1 \\ 0 & 0 & z \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \\
&= [1 \quad 1 \quad 1] \begin{bmatrix} (z-1)^{-1} & 0 & 0 \\ 0 & z^{-1} & z^{-2} \\ 0 & 0 & z^{-1} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \\
&= [(z-1)^{-1} \quad z^{-1} \quad z^{-2} + z^{-1}] \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ z-1 & z+1 \\ 0 & z^2 \end{bmatrix}.
\end{aligned}$$

- EXERCISE 4.7.1 Consider the single-input system $(F, \mathbf{g})$. If the Jordan form of the matrix F has at least two miniblocks relative to the same eigenvalue λ , then λ is an eigenvalue of the non-reachable subsystem.

We can obtain different standard reachability forms by choosing different bases for X , under the requirement that each of them is composed of a basis for X^R complemented with a basis for the full X . By doing so, the resulting matrices $\bar{F}$ all exhibit the same block triangular structure, whereas the actual entries of such blocks can vary.

However, notice that the eigenvalues of both $\bar{F}_{11}$ and $\bar{F}_{22}$ are independent of the basis provided the basis is chosen according to the described criterion. In fact, $\bar{F}_{11}$ represents the restriction of the transformation yielded by F over the F -invariant subspace X^R , while $\bar{F}_{22}$ is the linear transformation yielded by F over the quotient X/X^R .

Finally, changing basis of X^R – which corresponds to different choices of the first ρ vectors – and of the quotient X/X^R – which correspond on the other side to different choices of the last $n - \rho$ vectors – induces similarity transformations over $\bar{F}_{11}$ and $\bar{F}_{22}$.

- EXERCISE 4.7.2 Let $\bar{\mathcal{B}} = (\bar{\mathbf{v}}_1, \dots, \bar{\mathbf{v}}_\rho, \bar{\mathbf{v}}_{\rho+1}, \dots, \bar{\mathbf{v}}_n)$ and $\hat{\mathcal{B}} = (\hat{\mathbf{v}}_1, \dots, \hat{\mathbf{v}}_\rho, \hat{\mathbf{v}}_{\rho+1}, \dots, \hat{\mathbf{v}}_n)$ two bases for the state space of a system, and assume that the first ρ vectors ($\rho < n$) are, in both cases, a basis for the reachable subsystem. Then
 - i) the matrix of change of basis T that allows to express the vectors of $\hat{\mathcal{B}}$ as linear combinations of the vectors of $\bar{\mathcal{B}}$ is block triangular

$$T = \begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix}, \quad T_{11} \in \mathbb{R}^{\rho \times \rho}$$

- ii) if $\bar{F} = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix}$ and $\hat{F} = \begin{bmatrix} \hat{F}_{11} & \hat{F}_{12} \\ 0 & \hat{F}_{22} \end{bmatrix}$ are the state update equations of the system, in the basis $\bar{\mathcal{B}}$ and $\hat{\mathcal{B}}$ respectively, then $\hat{F}_{11} = T_{11}^{-1}\bar{F}_{11}T_{11}$ and $\hat{F}_{22} = T_{22}^{-1}\bar{F}_{22}T_{22}$.

5.8 Further reachability criteria

The previous sections provided a condition for reachability in a linear system $\Sigma = (F, G, H)$, either discrete-time or continuous-time, that can be expressed as

$$\text{rank} \begin{bmatrix} G & FG & F^2G & \dots & F^{n-1}G \end{bmatrix} = n.$$

It is often useful to adopt alternative, equivalent, reachability criterion, which highlights the relation between reachability and the internal structure of Σ in other ways than the rank condition $\mathcal{R} = n$. In this regard, the most interesting criterion is the so-called Popov, Belevitch and Hautus (PBH) criterion, and its corollaries. Further criteria will be introduced in the next chapters in connection with specific problems.

By employing the PBH criterion, the reachability analysis of an n -dimensional system Σ is done by studying the rank of the corresponding *PBH matrix*

$$[F - zI_n \quad G] \tag{5.79}$$

as z varies over the complex field.

Before stating the criterion, we notice two simple facts. First of all, we notice that, for arbitrary matrices F and G , the matrix (5.79) is of full rank, equal to n , for any z that is not in the spectrum of F . Therefore, the only values $z \in \mathbb{C}$ for which it can, possibly, have rank less than n are the eigenvalues of the matrix F .

Secondly, the rank of (5.79) and the rank of the PBH matrix

$$[\bar{F} - zI_n \quad \bar{G}] \tag{5.80}$$

of any system $\bar{\Sigma} = (\bar{F}, \bar{G}, \bar{H}) = (T^{-1}FT, T^{-1}G, HT)$ that is algebraically equivalent to Σ , coincide, for all $z \in \mathbb{C}$. In fact, it holds true

$$[\bar{F} - zI_n \quad \bar{G}] = [T^{-1}FT - T^{-1}zI_nT \quad T^{-1}G] = T^{-1} [F - zI_n \quad G] \begin{bmatrix} T & 0 \\ 0 & I_m \end{bmatrix}$$

and therefore the matrices (5.79) and (5.80) only differ by invertible left and right factors.

Theorem 5.8.1 [PBH CRITERION] *Let $\Sigma = (F, G, H)$ be an n -dimensional, either continuous-time or discrete-time, system. Then Σ is reachable if and only if the matrix (5.79) is full rank for all $z \in \mathbb{C}$.*

If the system is not reachable, the rank of the matrix (5.79) evaluated at z is less than n if and only if z is an eigenvalue of the non-reachable subsystem of Σ .

PROOF Assume that for $\lambda \in \mathbb{C}$ the rank of the matrix $[F - \lambda I_n \quad G]$ is less than n . Then its rows are linearly dependent and there exists a non-zero vector $\mathbf{v} \in \mathbb{C}^n$ such that

$$\mathbf{v}^T [F - \lambda I_n \quad G] = \mathbf{0}. \tag{5.81}$$

By (7.30) it follows

$$\begin{aligned} \mathbf{v}^T F &= \lambda \mathbf{v}^T & \Rightarrow & \mathbf{v}^T FG = \mathbf{0} \\ \mathbf{v}^T F^2 &= \lambda^2 \mathbf{v}^T & \Rightarrow & \mathbf{v}^T F^2G = \mathbf{0} \\ &\vdots & & \vdots \\ \mathbf{v}^T F^{n-1} &= \lambda^{n-1} \mathbf{v}^T & \Rightarrow & \mathbf{v}^T F^{n-1}G = \mathbf{0}. \end{aligned}$$

And therefore also

$$\mathbf{v}^T [G \quad FG \quad F^2G \quad \dots \quad F^{n-1}G] = \mathbf{0},$$

thus the rows of the reachability matrix are not linearly independent, and the system is not reachable.

Conversely, assume that the pair (F, G) be non-reachable, and consider the *standard reachability form*

$$\bar{F} = \begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix}, \quad \bar{G} = \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix},$$

where $(\bar{F}_{11}, \bar{G}_1)$ is reachable. Then, for all $\lambda \in \mathbb{C}$, it holds true

$$\begin{aligned} \text{rank} [F - \lambda I_n \quad G] &= \text{rank} [\bar{F} - \lambda I_n \quad \bar{G}] \\ &= \text{rank} \begin{bmatrix} \bar{F}_{11} - \lambda I_\rho & \bar{F}_{12} & \bar{G}_1 \\ 0 & \bar{F}_{22} - \lambda I_{n-\rho} & 0 \end{bmatrix} \end{aligned} \quad (5.82)$$

If $\lambda \in \Lambda(\bar{F}_{22})$, the matrix $\bar{F}_{22} - \lambda I_{n-\rho}$ is singular, and therefore the last $n - \rho$ rows of the PBH matrix of the pair $(\bar{F}, \bar{G})$ are linearly dependent. Therefore, the whole set of n rows of such a PBH matrix are linearly dependent, and its rank is thus not full. Consequently by (5.82), the rank of the PBH matrix of the pair (F, G) is also not full.

Finally, assume that there exists $\lambda \in \mathbb{C}$ such that $[F - \lambda I_n \quad G]$, and thus also $[\bar{F} - \lambda I_n \quad \bar{G}]$, is not full rank. Therefore, there exists a nonzero vector $\mathbf{v}^T = [\mathbf{v}_1^T \quad \mathbf{v}_2^T] \in \mathbb{C}^n$ for which it holds true

$$\mathbf{0} = [\mathbf{v}_1^T \quad \mathbf{v}_2^T] \begin{bmatrix} \bar{F}_{11} - \lambda I_\rho & \bar{F}_{12} & \bar{G}_1 \\ 0 & \bar{F}_{22} - \lambda I_{n-\rho} & 0 \end{bmatrix}$$

and therefore

$$\begin{aligned} \mathbf{0} &= \mathbf{v}_1^T [\bar{F}_{11} - \lambda I_\rho \quad \bar{G}_1] \\ \mathbf{0} &= \mathbf{v}_1^T \bar{F}_{12} + \mathbf{v}_2^T (\bar{F}_{22} - \lambda I_{n-\rho}). \end{aligned} \quad (5.83)$$

Since the pair $(\bar{F}_{11}, \bar{G}_1)$ is reachable, by the first part of the theorem it follows that the matrix $[\bar{F}_{11} - z I_\rho \quad \bar{G}_1]$ is full rank for all z , and thus $\mathbf{v}_1^T$ is zero. This implies that the second equation in (5.83) can be rewritten as

$$\mathbf{0} = \mathbf{v}_2^T (\bar{F}_{22} - \lambda I_{n-\rho}), \quad \mathbf{v}_2^T \neq \mathbf{0},$$

which can only hold true if $\bar{F}_{22} - \lambda I_{n-\rho}$ is singular, namely, if $\lambda \in \Lambda(\bar{F}_{22})$. ■

- EXERCISE 4.8.1 Assume $\text{rank} [G \quad FG \quad \dots \quad F^{n-1}G] < n$. Then

- i) there exists a nonzero vector $\mathbf{v}^T \in \mathbb{R}^n$ such that $\mathbf{v}^T [G \quad FG \quad \dots \quad F^{n-1}G] = \mathbf{0}$;
- ii) if $\psi_{\mathbf{v}}(s)$ is the minimum annihilating polynomial of $\mathbf{v}$ with respect to F^T and if λ is a zero of $\psi_{\mathbf{v}}(s)$, there exists a polynomial $g(s) = \gamma_0 + \gamma_1 s + \dots + s^\nu$, with $\nu < \deg \psi_{\mathbf{v}}$, for which it holds true

$$\mathbf{0} = \mathbf{v}^T \psi_{\mathbf{v}}(F) = \mathbf{v}^T g(F)(F - \lambda I);$$

- iii) the row vector $\mathbf{w}^T = \mathbf{v}^T g(F)$ is not the zero vector, and satisfies the condition $\mathbf{w}^T G = \gamma_0 \mathbf{v}^T G + \gamma_1 \mathbf{v}^T FG + \dots + \mathbf{v}^T F^\nu G = \mathbf{0}$;
- iv) $\mathbf{w}^T [F - \lambda I_n \quad G] = \mathbf{0}$ and the PBH matrix has rank less than n for $z = \lambda$, thus providing an alternative proof of the second part of Theorem 5.8.1.

The previous theorem allows us to state and prove the next result, a reachability criterion which can be applied when F is in Jordan form.

Corollary 5.8.2 [JORDAN STRUCTURE OF REACHABLE SYSTEMS] *Consider a system $\Sigma = (F_J, G, H)$, with F_J in canonical Jordan form. Then Σ is reachable if and only if, for any eigenvalue λ_i of F_J , the rows in G corresponding to the last rows of the Jordan miniblocks relative to λ_i are linearly independent.*

PROOF Consider the PBH matrix for $z = \lambda_i \in \Lambda(F_J)$

$$[F_J - \lambda_i I_n \quad G]. \quad (5.84)$$

It is straightforward to verify that the last rows of the miniblock of $F_J - \lambda_i I_n$ that correspond to λ_i are zero. Therefore, (5.84) can be full rank only if the corresponding rows of G are linearly independent. On the other side, if this holds true, it is easy to verify that all rows of (5.84) are linearly independent.

This concludes the proof, since (F_J, G) is reachable if and only if the PBH matrix is of full rank for all eigenvalues of F_J . ■

If a system $\Sigma = (F, G, H)$ has m inputs, the reachability condition provided in the corollary is not compatible with the existence of more than m Jordan miniblocks corresponding to the same eigenvalue in the Jordan form of F , namely, prohibits any eigenvalue to have geometric multiplicity greater than m .

In the particular case of a single input system $\Sigma = (F, \mathbf{g}, H)$, reachability implies that the Jordan form F_J of F only has a single miniblock for each eigenvalue, namely, that F is cyclic.

Since the change of basis that reduces F into its Jordan form F_J is not unique, it is possible to exploit such a degree of freedom to change the column $\mathbf{g}$ into a particularly simple form. Indeed, there exists a change of basis that transforms $(F, \mathbf{g})$ into the pair $(F_J, \mathbf{g}_J)$ described as

$$F_J = \begin{bmatrix} \lambda_1 & 1 & & & \\ & \ddots & \ddots & & \\ & & \lambda_1 & 1 & \\ & & & \lambda_1 & 1 \\ \hline & & & & \ddots & \ddots & \\ & & & & & \ddots & \ddots & \\ & & & & & & \lambda_t & 1 \\ & & & & & & & \lambda_t & 1 \\ & & & & & & & & \lambda_t \end{bmatrix}, \quad \mathbf{g}_J = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \hline \vdots \\ \hline 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}. \quad (5.85)$$

In order to show this, assume that a first change of basis has transformed the reachable pair $(F, \mathbf{g})$ into a pair $(F_J, \tilde{\mathbf{g}})$ with F_J in Jordan form. In the new pair the vector $\tilde{\mathbf{g}}$ has non-zero entries corresponding to the last row of each Jordan miniblock. Assume now that $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(\nu)}$ is the Jordan chain corresponding to one of the miniblocks. Then, for

$\alpha_\nu \neq 0$ and arbitrary $\alpha_1, \dots, \alpha_{\nu-1}$, the vector sequence $\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(\nu)}$ defined by

$$\begin{aligned} \mathbf{w}^{(1)} &= \alpha_\nu \mathbf{v}^{(1)} \\ \mathbf{w}^{(2)} &= \alpha_\nu \mathbf{v}^{(2)} + \alpha_{\nu-1} \mathbf{v}^{(1)} \\ &\vdots \\ \mathbf{w}^{(\nu-1)} &= \alpha_\nu \mathbf{v}^{(\nu-1)} + \alpha_{\nu-1} \mathbf{v}^{(\nu-2)} + \dots + \alpha_2 \mathbf{v}^{(1)} \\ \mathbf{w}^{(\nu)} &= \alpha_\nu \mathbf{v}^{(\nu)} + \alpha_{\nu-1} \mathbf{v}^{(\nu-1)} + \dots + \alpha_2 \mathbf{v}^{(2)} + \alpha_1 \mathbf{v}^{(1)}, \end{aligned}$$

is also a Jordan chain that generates the same subspace of $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(\nu)}$. $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(\nu)}$. The miniblock corresponding to $\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(\nu)}$ then coincides with the one associated to $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(\nu)}$, and the matrix of change of basis from $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(\nu)}$ into $\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(\nu)}$ is the Toeplitz matrix⁶:

$$T = \begin{bmatrix} \alpha_\nu & \alpha_{\nu-1} & \alpha_{\nu-2} & \cdots & \alpha_1 \\ & \alpha_\nu & \alpha_{\nu-1} & \cdots & \alpha_2 \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & \alpha_{\nu-1} \\ & & & & \alpha_\nu \end{bmatrix}. \quad (5.86)$$

Notice now that the set of upper triangular Toeplitz matrices with non-zero diagonal entries is closed with respect to multiplication, contains the identity, and also the inverse of any element of the set. Therefore, it is a multiplicative group.

For any ν -ple of $\mathbb{C}^\nu$ whose last entry is non-zero, there exists in the group a matrix T^{-1} that reduces it to the canonical ν -ple $\mathbf{e}_\nu$. It is then straightforward to verify that there exists a suitable block diagonal matrix, whose diagonal Toeplitz blocks have the same dimensions as the miniblocks in F_J and structure (5.86), that induces a change of basis that does not modify the structure of F_J and transforms $\tilde{\mathbf{g}}$ into $\mathbf{g}_J$.

- EXERCISE 4.8.2 [ALTERNATIVE PROOF OF THE STRUCTURE OF T] (i) If $J = \lambda I_n + N$ is a Jordan miniblock, the equation $JT = TJ$ in the matrix unknown T is equivalent to $0 = TN - NT$. The latter has as solution an arbitrary upper triangular Toeplitz matrix whose diagonal is non-zero if T is invertible.
- ii) Assume $J_i = \lambda_i I + N_i$, $i = 1, 2$, are Jordan miniblocks of dimension $\nu_i \times \nu_i$ corresponding to eigenvalues $\lambda_1 \neq \lambda_2$, and consider the Jordan form $J = \text{diag}\{J_1, J_2\}$. The equation $JT = TJ$ in the unknown

$$T = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}$$

is equivalent to the system of equations

$$\begin{aligned} (\lambda_1 - \lambda_2)T_{12} &= N_1 T_{12} - T_{12} N_2 \\ (\lambda_1 - \lambda_2)T_{21} &= N_2 T_{21} - T_{21} N_1 \\ 0 &= N_1 T_{11} - T_{11} N_1 \\ 0 &= N_2 T_{22} - T_{22} N_2. \end{aligned}$$

Verify that both T_{12} and T_{21} are zero, and thus, by (i), the solutions to the system of equations are the block diagonal matrices $T = \text{diag}\{T_{11}, T_{22}\}$, where the blocks T_{ii} are upper triangular Toeplitz matrices, whose diagonal entries are non-zero if T is also required to be invertible.

- iii) Extend the previous result to an arbitrary Jordan form of a cyclic matrix.

⁶A matrix $T = [t_{i,j}] \in \mathbb{C}^{\nu \times \nu}$ is said to be Toeplitz if, for any $i, j < \nu$, it holds true $t_{i,j} = t_{i+1,j+1}$

5.9 Bibliography

The structural properties of linear continuous-time dynamical systems are thoroughly discussed in

1. J.C.Willems, S.K.Mitter “*Controllability, observability, pole allocation and state reconstruction*” IEEE Trans. on Automatic Control, vol AC-16, pp.582-95, 1971

For Gramian matrices over infinite horizon, see

2. A.Antoulas “*Approximation of large scale dynamical systems*” SIAM, 2009
3. B. C. Moore “*Principal component analysis in linear systems: Controllability, observability, and model reduction*” IEEE Trans. on Automatic Control, vol. AC-26, pp. 17-32, 1981.

We quickly hinted at the problem of movement over subspaces, a topic that belongs to the so called “Geometric Theory” of linear systems. For a complete overview, we refer to the monographs of the two scholars who started such a research branch:

4. G.Marro “*Teoria dei sistemi e del controllo*” Zanichelli, 2000
5. W.M.Wonham “*Linear multivariable control: a geometric approach*”, Springer-Verlag, 1985

The PBH reachability criterion (and the corresponding observability criterion) and its consequences are an important instance of how problems related to analysis and control of linear systems can be studied by means of matricial polynomials. For a more in-depth analysis:

6. V.Kucera “*Discrete linear control: the polynomial approach*” Academia (Praga), 1979
7. F.M.Callier, C.A.Desoer “*Multivariable feedback systems*” Springer-Verlag, 1982